



Project Report On

Edge-Enabled Drone for Disaster Response

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Computer Science and Engineering

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CERTIFICATE

*This is to certify that the project report entitled "**Edge-Enabled Drone for Disaster Response**" is a bonafide record of the work done by **Niranjana Vijay K (U2203164)**, **Sandra Anna Priyan(U2201181)**, **Sreelakshmi K S (U2203197)** and **Teryn Thomas P (U2203203)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

Rapid detection and localization of victims is a major challenge during disaster response operations, particularly when communication infrastructure is damaged and affected areas become inaccessible to rescue teams. This project proposes an edge-enabled drone-based disaster response system designed to assist rescue teams in detecting and locating victims efficiently in disaster-affected environments. The system integrates Edge AI-based visual detection with passive mobile signal detection to improve the reliability of victim identification. A YOLO based deep learning model processes images captured by the drone camera to detect human presence, while a Passive Mobile Detection Module scans for nearby mobile devices by capturing Bluetooth signals to estimate the location of individuals even when they are not directly visible. Detection results are transmitted to the ground station using LoRa-based communication, enabling long-range data transfer even in areas without conventional network connectivity. The proposed system is implemented using a combination of drone simulation and a Raspberry Pi-based hardware prototype to demonstrate the feasibility of detection, localization, and communication technologies. The system highlights the potential of combining Edge AI, passive signal detection, and LoRa communication to support faster and more efficient search and rescue operations in disaster scenarios.

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List of Abbreviations

AI	Artificial Intelligence
GPS	Global Positioning System
LoRa	Long Range Radio Communication
MAC	Media Access Control
RSSI	Received Signal Strength Indicator
SAR	Search and Rescue
SITL	Software-In-The-Loop
UAV	Unmanned Aerial Vehicle
Wi-Fi	Wireless Fidelity
YOLO	You Only Look Once

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Chapter 1

Introduction

This chapter provides an overview of the Edge-Enabled Drone for Disaster Response system, outlining the purpose, scope, and significance of the proposed solution. It begins with a discussion of the background of disaster response operations and the limitations of traditional search and rescue methods. The chapter then presents the problem definition, scope, and motivation for developing an intelligent drone-based system for victim detection. It further describes the objectives, challenges, and assumptions associated with the proposed system. Finally, the societal and industrial relevance of the project is discussed, followed by an outline of the overall organization of the report.

1.1 Background

Natural disasters such as floods, earthquakes, landslides, and cyclones often cause severe damage to infrastructure, making communication networks and transportation systems unreliable or completely unavailable. During the initial hours following a disaster, rapid identification and localization of survivors becomes critical for minimizing casualties. Traditional search and rescue operations rely heavily on manual ground surveys, which are time-consuming, labor-intensive, and risky for rescue personnel.

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have emerged as promising tools for disaster management due to their ability to quickly survey large areas. Recent advancements in edge computing and lightweight artificial intelligence models enable drones to perform real-time data processing onboard without relying on continuous internet connectivity. Additionally, passive mobile signal detection techniques allow identification of nearby personal devices by analyzing wireless signal strength, even in communication-blackout scenarios.

However, most existing drone-based systems depend on centralized cloud infrastruc-

ture for data processing and coordination. In disaster-affected environments, such dependencies can significantly reduce system effectiveness. Therefore, there is a growing need for autonomous, low-cost, and offline-capable drone systems that can operate reliably in connectivity-deprived conditions.

1.2 Problem Definition

During disaster situations, locating victims quickly is difficult due to debris, inaccessible areas, and damaged infrastructure. Traditional search and rescue methods are slow and require significant manpower, while communication networks are often disrupted. Therefore, there is a need for an autonomous system that can efficiently detect and locate victims and transmit the information to rescue teams even in environments without reliable communication infrastructure.

1.3 Scope and Motivation

1.3.1 Scope

The scope of this project is to design and demonstrate an edge-enabled drone-based disaster response system capable of detecting and locating victims in disaster-affected areas. The system integrates Edge AI-based visual detection with passive mobile signal detection to improve victim identification. RSSI-based distance estimation and trilateration are used to estimate the victim's location, while LoRa communication enables transmission of detection information to the ground station even in environments where conventional communication networks are unavailable. The system is validated using drone simulation and a Raspberry Pi-based prototype. Drone behavior and navigation are tested using simulation tools such as ArduPilot SITL and MAVProxy. A Raspberry Pi-based prototype is developed to demonstrate real-time signal scanning, GPS coordinate logging, and localization capability. Additionally, a ground station dashboard is implemented to visualize drone path, heatmaps, and estimated victim coordinates for rescue team support.

1.3.2 Motivation

The motivation for this project arises from the limitations of traditional disaster response methods, which rely heavily on manual search operations and often face delays in locating

victims. In disaster environments where infrastructure is damaged and communication networks are disrupted, rapid detection and reliable information transmission become critical. Advances in drone technology, edge computing, and wireless communication provide an opportunity to develop intelligent systems that can assist rescue teams in locating victims faster and improving the efficiency of search and rescue operations.

1.4 Objectives

- To design a drone-based disaster response system capable of detecting and locating victims in disaster-affected environments.
- To implement Edge AI-based human detection using deep learning techniques for analyzing aerial images captured by the drone.
- To perform passive mobile signal detection by scanning wireless signals emitted by nearby mobile devices.
- To estimate the distance between the drone and detected devices using RSSI-based signal strength measurements.
- To determine the approximate location of victims using a trilateration algorithm.
- To transmit victim detection information to the ground station using LoRa-based long-range communication.
- To create a ground station dashboard for real-time visualization of drone path, detections, and victim coordinates.
- To demonstrate the feasibility of the proposed system through drone simulation and a Raspberry Pi-based hardware prototype.

1.5 Challenges

The development of a drone-based disaster response system involves several technical challenges:

- **Environmental Constraints:** Visual victim detection may be affected by poor lighting conditions, debris, and obstacles commonly present in disaster areas.

- **Signal Interference:** RSSI-based distance estimation used for victim localization can be affected by signal interference, multipath propagation, obstacles, and environmental noise.
- **Limited Hardware Resources:** Edge devices such as Raspberry Pi have limited processing capability, making it challenging to run AI models and signal processing algorithms efficiently.
- **System Integration:** Integrating multiple modules including visual detection, signal scanning, localization, and communication while maintaining stable system performance is a significant design challenge.

1.6 Assumptions

- Victims are carrying mobile devices that emit detectable Wi-Fi or Bluetooth signals.
- The drone has access to approximate GPS coordinates for logging movement during scanning.
- Environmental interference does not completely block wireless signal detection.
- The prototype operates within a controlled test environment for validation purposes.

1.7 Societal and Industrial Relevance

This project has significant relevance for both society and industry by improving the efficiency and reliability of disaster response operations through the integration of advanced technologies.

- **Societal Impact:** The system assists rescue teams in locating victims faster, potentially saving lives during disaster situations.
- **Improved Disaster Management:** Automated victim detection and localization can significantly reduce search time and enhance emergency response efficiency.
- **Industrial Applicability:** The proposed system can be used by disaster management agencies, emergency services, and humanitarian organizations for monitoring and rescue operations.

- **Technological Advancement:** The project demonstrates the integration of UAV technology, edge computing, artificial intelligence, and wireless communication in a unified disaster response framework.
- **Scalability and Future Applications:** The system architecture can be extended to support multi-drone coordination, environmental monitoring, and smart city disaster management systems.

1.8 Organization of the Report

This report is structured as follows:

- **Chapter 1: Introduction** presents the background, problem definition, scope, objectives, challenges, and relevance of the proposed system.
- **Chapter 2: Literature Review** analyzes previous research related to UAV-based disaster response systems, victim detection methods, and communication technologies.
- **Chapter 3: System Design** describes the overall system architecture, design components, module division, tools and technologies used, and project workflow.
- **Chapter 4: Implementation Scope** explains the implementation approach using simulation and hardware prototype development.
- **Chapter 5: Results and Discussion** presents the experimental results and analysis of the developed system.
- **Chapter 6: Conclusions and Future Scope** summarizes the findings of the project and outlines potential improvements for future work.

1.9 Summary of Chapter

This chapter introduced the Edge-Enabled Drone for Disaster Response system and discussed the motivation for developing an intelligent victim detection framework. The chapter highlighted the limitations of traditional rescue operations and explained how the integration of Edge AI, passive mobile signal detection, and LoRa communication can

improve disaster response capabilities. The problem definition, objectives, challenges, and assumptions associated with the system were discussed, along with the societal and industrial relevance of the project. Finally, the organization of the report was presented.

Chapter 2

Literature Review

2.1 Introduction

Recent advancements in unmanned aerial vehicles (UAVs), artificial intelligence, and wireless communication technologies have significantly improved disaster response systems. UAV-based solutions can provide rapid aerial surveillance, real-time victim detection, and improved situational awareness during search and rescue operations. Researchers have explored different techniques such as computer vision-based human detection, wireless signal localization, and autonomous UAV systems to improve disaster management.

This chapter reviews several existing research works related to UAV-assisted search and rescue systems, edge AI-based victim detection, and wireless signal localization techniques. The objective of this review is to analyze the strengths and limitations of existing approaches and identify research gaps that motivate the development of the proposed edge-enabled drone-based disaster response system.

2.2 Existing Techniques

2.2.1 Edge AI Based Autonomous UAV for Emergency Network Deployment: A Study Towards Search and Rescue Missions [1]

Problem Addressed

In many disaster scenarios, communication infrastructure may be damaged, preventing victims from contacting rescue teams. UAVs equipped with edge computing and communication capabilities can assist by detecting survivors and providing temporary communication networks.

Objective

The objective of this work is to develop an edge-enabled UAV system capable of performing real-time survivor detection while simultaneously deploying emergency communication networks.

Methodology

- **Aerial Monitoring:** UAVs are deployed over disaster areas to capture aerial images and video streams of the environment. These images provide a comprehensive view of affected regions and assist rescue teams in identifying potential victim locations.
- **Search Strategy Planning:** An intelligent search strategy is implemented to guide UAV navigation. Probabilistic models are used to prioritize areas where victims are most likely to be located based on environmental information.
- **Autonomous Navigation:** The UAV autonomously navigates through predefined search paths while avoiding obstacles and maintaining stable flight conditions.
- **Data Transmission:** Captured data is transmitted in real time to ground control stations where rescue operators analyze the information and coordinate response efforts.
- **Victim Identification Support:** Rescue teams use the aerial imagery to identify signs of human presence such as movement, clothing, or distress signals.

Advantages

Edge processing reduces latency and enables real-time decision making without relying on cloud connectivity.

Disadvantages

Edge devices have limited computational capacity, and UAV flight duration remains constrained by battery limitations.

2.2.2 Supporting Search and Rescue Operations with UAVs [2]

Problem Addressed

In disaster situations such as earthquakes or building collapses, identifying victims quickly is essential for successful rescue operations. Traditional search and rescue teams often face challenges when navigating hazardous environments, especially when infrastructure is damaged or when access to certain locations is restricted. Manual search methods can be slow and may expose rescuers to significant risks.

The study by Waharte and Trigoni investigates how UAVs can assist in search and rescue missions by providing aerial monitoring and intelligent search strategies. The objective is to improve the efficiency of rescue operations by enabling faster identification of victims and better coordination between rescue teams.

Objective

The primary objective of this work is to develop a UAV-assisted system capable of supporting search and rescue operations through aerial monitoring and autonomous navigation. The system aims to efficiently explore disaster areas, detect possible victims, and transmit real-time information to rescue teams on the ground.

Methodology

The proposed approach focuses on supporting search and rescue teams through systematic aerial exploration and monitoring of disaster areas using UAVs.

- **Area Mapping:** The UAV first performs an aerial survey of the disaster region to generate a map of the affected environment. This helps identify damaged structures, blocked roads, and inaccessible areas.
- **Grid-Based Search:** The disaster area is divided into multiple search grids. The UAV follows a structured path to scan each grid systematically, ensuring that the entire region is covered without missing potential victim locations.
- **Priority Region Identification:** Probabilistic models are used to determine areas where victims are more likely to be present, such as near collapsed buildings or open spaces. The UAV prioritizes these regions during the search process.

- **Visual Monitoring:** The onboard camera continuously captures images and video streams while the UAV is in flight. These visuals help identify possible signs of human presence in the disaster zone.
- **Ground Team Support:** The collected aerial data is transmitted to rescue teams to assist them in planning safe entry routes, locating victims, and coordinating rescue operations effectively.

Advantages

The UAV-based monitoring system enables rapid coverage of large disaster areas and improves situational awareness for rescue teams. It significantly reduces the time required to locate victims and minimizes risks faced by human rescuers.

Disadvantages

The system is limited by UAV battery capacity, which restricts flight duration. Additionally, environmental conditions such as strong winds or poor visibility may affect UAV performance and data quality.

2.2.3 Unsupervised Human Detection with an Embedded Vision System on a Fully Autonomous UAV for Search and Rescue Operations [3]

Problem Addressed

Detecting human survivors in disaster environments using aerial imagery is a challenging task due to varying environmental conditions and the complexity of visual scenes. Manual monitoring of UAV video feeds can be inefficient and prone to human error.

The study by Lygouras et al. proposes the use of embedded vision systems integrated within UAV platforms to automatically detect humans in aerial images.

Objective

The objective of this research is to develop an automated human detection system that operates directly on UAV hardware using embedded vision technologies. The system aims to perform real-time analysis of aerial images and identify potential survivors without requiring constant human monitoring.

Methodology

The system integrates computer vision techniques with embedded processing hardware installed on the UAV. The key steps include:

- **Image Acquisition:** The UAV captures aerial images using onboard cameras while flying over the disaster region.
- **Preprocessing:** Captured images are processed to enhance contrast, remove noise, and prepare them for further analysis.
- **Feature Extraction:** Relevant visual features such as edges, shapes, and textures are extracted from the images.
- **Human Detection Algorithm:** Computer vision algorithms analyze the extracted features to identify patterns corresponding to human shapes.
- **Onboard Processing:** The detection algorithm runs on embedded processors within the UAV, allowing real-time analysis without requiring remote servers.
- **Detection Reporting:** When a potential human target is detected, the UAV transmits the location and image data to rescue teams.

Advantages

The embedded vision system enables automatic detection of victims and reduces reliance on human operators. Onboard processing allows faster response times since image analysis is performed locally.

Disadvantages

Detection accuracy may be affected by lighting conditions, shadows, or obstacles such as debris. Additionally, embedded processors on UAVs have limited computational resources compared to powerful ground-based systems.

2.2.4 Localization of WiFi Devices Using Probe Requests Captured at Unmanned Aerial Vehicles [4]

Problem Addressed

In disaster scenarios, many victims may carry mobile devices that periodically transmit WiFi probe requests while searching for available networks. These signals can potentially be used to detect and locate survivors even if they are not visible in aerial imagery.

The work by Acuna et al. explores how UAVs can capture these wireless signals and estimate the location of victims through signal strength measurements.

Objective

The objective of this research is to develop a UAV-based system capable of detecting WiFi probe requests from mobile devices and estimating their geographic location using signal analysis techniques.

Methodology

The proposed system combines wireless signal monitoring with localization algorithms. The process includes the following steps:

- **Signal Detection:** The UAV is equipped with wireless receivers that monitor WiFi probe requests emitted by nearby devices.
- **Signal Strength Measurement:** The Received Signal Strength Indicator (RSSI) of detected probe requests is measured.
- **Distance Estimation:** Using RSSI values, the approximate distance between the UAV and the transmitting device is estimated.
- **Position Estimation:** Localization algorithms combine multiple RSSI measurements from different UAV positions to estimate the probable location of the device.
- **Victim Localization:** The estimated coordinates are transmitted to rescue teams to guide search efforts.

Advantages

The approach allows victims to be detected through their mobile devices even when they are hidden under debris or not visible in images. UAVs can cover large areas quickly and collect wireless data efficiently.

Disadvantages

Localization accuracy may be affected by signal interference, obstacles, or multipath effects. Additionally, the method requires victims to possess WiFi-enabled devices.

2.2.5 UAV-based Real-Time Survivor Detection System in Post-disaster Search and Rescue Operations[5]

Problem Addressed

Manual analysis of UAV imagery for survivor detection is time-consuming and inefficient during large-scale disasters. Automated object detection systems based on deep learning can significantly improve detection speed and accuracy.

Objective

The objective of this research is to develop a UAV-based real-time survivor detection system using deep learning techniques capable of identifying human victims in aerial imagery. The system aims to utilize UAV-mounted cameras to capture images of disaster-affected areas and apply advanced object detection algorithms to automatically recognize human presence. By integrating aerial imaging with deep learning models, the system seeks to improve the speed and accuracy of victim identification. This approach supports rescue teams by providing timely information about possible survivor locations, thereby enhancing the efficiency of search and rescue operations in disaster scenarios.

Methodology

The system integrates UAV imaging with a deep learning-based object detection model.

- **Image Capture:** UAVs capture continuous aerial images of the disaster area.

- **Image Transmission:** Captured images are transmitted to processing units for analysis.
- **Deep Learning Detection:** A YOLO-based object detection model analyzes the images to identify human figures.
- **Bounding Box Generation:** Detected survivors are highlighted using bounding boxes indicating their positions.
- **Location Reporting:** Detected locations are transmitted to rescue teams for immediate action.

Advantages

The system enables real-time detection of survivors and improves the efficiency of search operations by automating the identification process.

Disadvantages

Deep learning models require significant computational resources and may struggle in scenarios with occlusion, debris, or complex backgrounds.

2.3 Comparison of Related Works

Paper	Technique Used	Key Contribution	Limitations
Edge AI Based Autonomous UAV for Emergency Network Deployment: A Study Towards Search and Rescue Missions [1]	Edge AI-based UAV with onboard processing and emergency network deployment	Enables real-time survivor detection using Edge AI and supports emergency communication deployment for disaster environments	Implementation complexity and limited onboard computing resources
Supporting Search and Rescue Operations with UAVs [2]	UAV-based aerial monitoring and intelligent search strategies	Demonstrates how UAVs can support rescue teams by covering large disaster areas and providing aerial situational awareness	Does not include automated victim detection or localization methods
Unsupervised Human Detection with an Embedded Vision System on a Fully Autonomous UAV for Search and Rescue Operations [3]	Embedded vision system for human detection on UAV	Enables onboard human detection using computer vision techniques on embedded hardware	Detection accuracy depends heavily on lighting conditions and environmental factors
Localization of WiFi Devices Using Probe Requests Captured at Unmanned Aerial Vehicles [4]	WiFi probe request capture and RSSI-based localization	Uses UAVs to detect mobile devices and estimate their location using signal strength measurements	Localization accuracy affected by signal interference and environmental noise
UAV-Based Real-Time Survivor Detection System in Post-Disaster Search and Rescue Operations [5]	Deep learning based human detection using UAV imagery	Applies YOLO-based deep learning models for automatic survivor detection from aerial images	Requires significant computational resources for real-time processing

2.3.1 Gaps Identified

From the comparison of existing studies, it can be observed that many UAV-based disaster response systems rely on a single victim detection technique, such as computer vision-based human detection or wireless signal-based localization. While these approaches provide useful information, relying on a single method may reduce reliability in complex disaster environments where factors such as poor lighting, obstacles, or signal interference can affect detection performance.

Several systems also assume the availability of stable communication infrastructure or centralized processing, which may not be available in real disaster scenarios. Damage to cellular or internet infrastructure can limit the ability of UAV systems to transmit detection results to rescue teams in real time.

The work by Biswas et al. [1] demonstrates the advantages of Edge AI-based UAV systems that perform onboard processing for disaster response. However, there remains a need to improve system reliability by integrating multiple detection methods and robust communication mechanisms.

2.4 Summary of Chapter

This chapter reviewed existing research on UAV-based disaster response systems, focusing on methods for victim detection, localization, and communication. The studies highlight the use of aerial monitoring, computer vision-based human detection, and wireless signal analysis to support search and rescue operations. These approaches demonstrate the potential of UAVs to improve situational awareness and search efficiency in disaster environments.

However, the review also identifies several limitations, including reliance on single detection techniques and dependence on stable communication infrastructure. Such limitations may reduce system reliability in real disaster scenarios. These gaps motivate the proposed system, which integrates Edge AI-based visual detection, passive mobile signal detection using RSSI, and LoRa communication to improve the reliability and effectiveness of victim detection and information transmission.

Chapter 3

System Design

This chapter presents the design and structure of the proposed drone-based disaster response system. The system is designed to assist rescue teams in locating victims in disaster-affected areas using a combination of Edge AI-based visual detection and passive mobile signal detection. The architecture also incorporates LoRa-based communication to enable long-range transmission of victim information to the ground station in environments where conventional communication networks may be unavailable. The overall architecture, component design, module division, tools used, and project workflow are described.

3.1 Overall Architecture

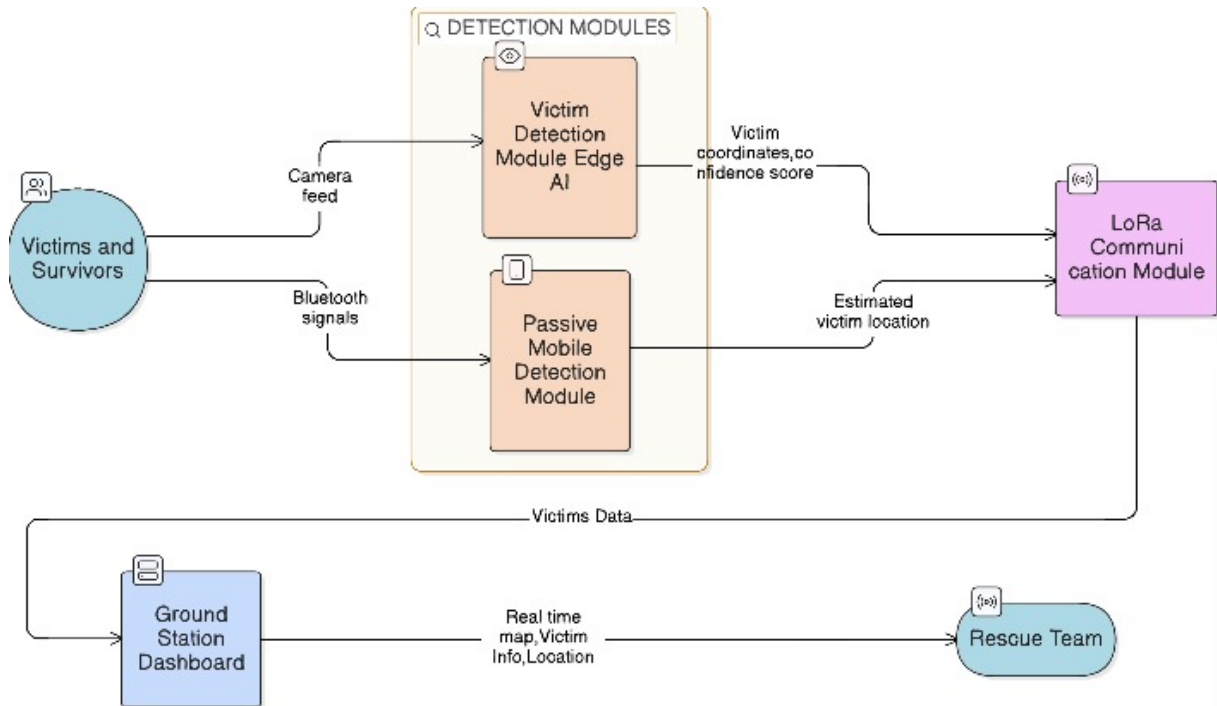


Figure 3.1: System Architecture

The overall architecture of the proposed disaster response system illustrates how different components interact to detect and locate victims in a disaster environment. The system is designed with a modular structure so that each functional component performs a specific task while contributing to the overall operation of the system.

The architecture consists of four major components: the drone system, detection modules, communication module, and ground station. The drone system operates as the primary sensing platform that monitors the affected environment and collects visual and wireless signal data. These data sources are processed by dedicated detection modules onboard the drone.

The Edge AI-based victim detection module analyzes images captured by the drone camera to identify the presence of humans. This module uses computer vision techniques to detect potential victims and generate information such as detection confidence and approximate coordinates.

The passive mobile signal detection module complements the visual detection system by identifying wireless signals emitted from nearby mobile devices. By measuring the received signal strength indicator (RSSI) values from detected devices, the system can estimate the proximity of potential victims even when they are not directly visible.

The processed victim information is transmitted to the ground station through the LoRa communication module. The ground station acts as the control interface where rescue teams can monitor detected victim locations, visualize system information, and coordinate rescue operations. This modular architecture enables reliable victim detection, localization, and communication in disaster environments.

3.2 Component Design

3.2.1 Sequence Diagram

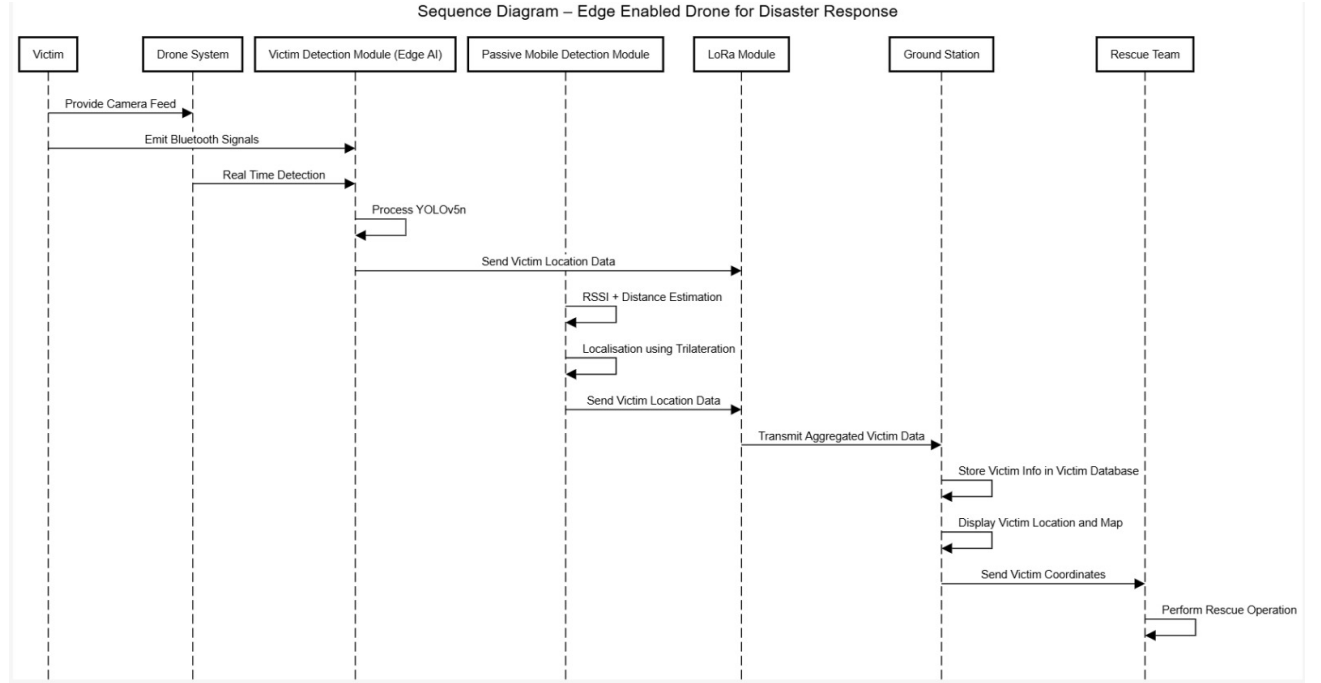


Figure 3.2: Sequence Diagram

This diagram illustrates the sequence of interactions between the drone system, edge AI modules, communication modules, and the rescue team during the victim detection and disaster response process. It demonstrates how the UAV captures aerial images of the disaster area and processes them using edge AI models to detect possible survivors. Once a potential victim is identified, the system communicates the detection results and location information through the communication module. The transmitted data is then received by the rescue team, enabling them to quickly analyze the situation and coordinate appropriate rescue actions. This interaction ensures efficient information flow between the UAV system and rescue personnel, improving the speed and effectiveness of disaster response operations.

- **Victim Detection Flow:** The victim is captured through the UAV camera system, and the live camera feed is received by the drone system. The Victim Detection Module (Edge AI) processes the incoming images in real time using the YOLOv5n

model. The AI model analyzes the camera stream, detects possible human victims, and extracts relevant detection information for further processing.

- **Mobile Signal Detection Flow:** The victim's mobile device emits wireless signals such as Bluetooth. These signals are captured by the Passive Mobile Detection Module mounted on the drone. The received signal strength indicator (RSSI) values are analyzed to estimate the distance between the drone and the mobile device. Using multiple measurements, trilateration is applied to estimate the approximate location of the victim.
- **Communication and Data Transmission Flow:** Once the victim is detected and the approximate location is determined, the victim data is forwarded to the LoRa communication module. The LoRa module aggregates the detection information and transmits the victim location data to the ground station over a long-range communication link.
- **Ground Station Processing Flow:** The ground station receives the transmitted victim information from the UAV system. The received data is stored in the victim database for further analysis and record keeping. The system then visualizes the victim locations on a map-based interface to help operators monitor the disaster area effectively.
- **Rescue Operation Flow:** The ground station forwards the detected victim coordinates to the rescue team. Using this location information, the rescue team can quickly reach the identified area and carry out rescue operations efficiently.

3.2.2 Use Case Diagram

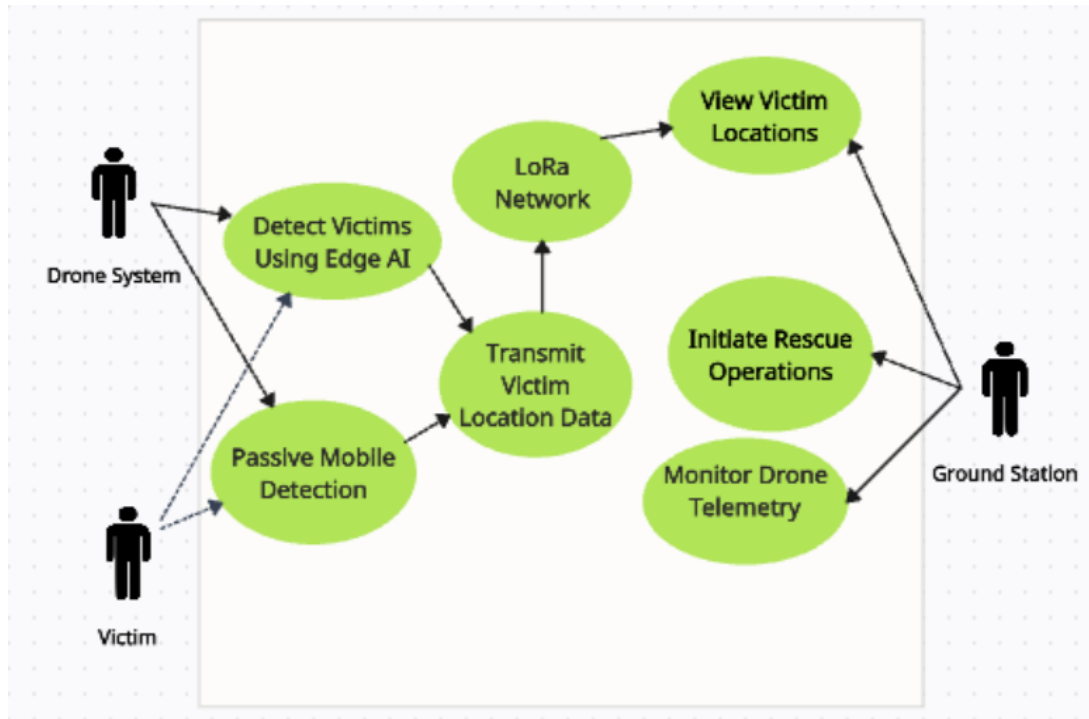


Figure 3.3: Use Case Diagram

The use case diagram illustrates the interaction between the proposed disaster response system and the external entities involved in the rescue operation. It identifies the main actors of the system and the primary activities performed to detect and locate victims in disaster-affected areas.

Actors:

- **Ground Station Operator:** Monitors the drone system, views detected victim locations, and initiates rescue [operations through the ground station interface.
- **Drone System:** Performs aerial monitoring of the disaster area, detects potential victims using onboard detection modules, and communicates collected information to the ground station.
- **Victim:** Represents individuals present in the disaster area whose presence may be detected either visually or through signals emitted by their mobile devices.

Primary Use Cases:

1. **Victim Detection:** The drone identifies potential victims using visual analysis and passive mobile signal detection while surveying the disaster area.
2. **Data Transmission:** Information related to detected victims and their estimated locations is transmitted from the drone to the ground station through the communication network.
3. **Location Visualization:** The ground station displays the detected victim locations and relevant information to assist rescue teams in decision making.

3.3 Module Division

3.3.1 Edge AI-Based Victim Detection Module

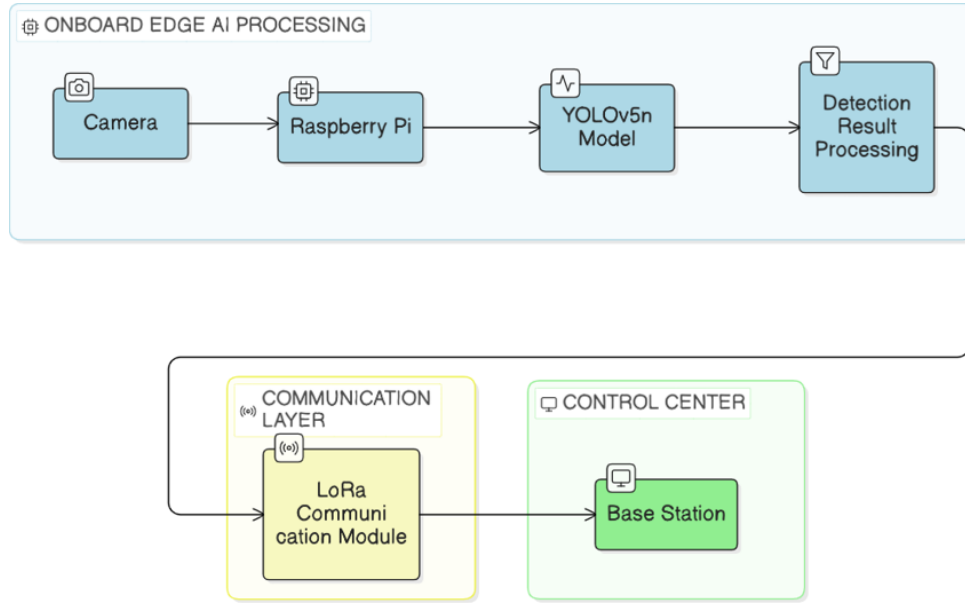


Figure 3.4: Edge AI Detection Module

The Edge AI-based victim detection module is responsible for identifying human presence in the disaster environment using visual data captured by the drone camera. The captured images are processed onboard using an embedded computing platform such as the Raspberry Pi.

The YOLOv5n deep learning model is used to perform real-time object detection on the incoming video stream. The model analyzes each frame and identifies potential victims by detecting human figures within the captured images. Once a detection occurs,

the module generates information such as bounding box coordinates, detection confidence score, and timestamp. This information is then forwarded to the communication module for further processing and transmission.

By performing inference directly on the edge device, the system reduces dependence on cloud connectivity and enables faster response times in disaster environments.

3.3.2 Passive Mobile Signal Detection Module

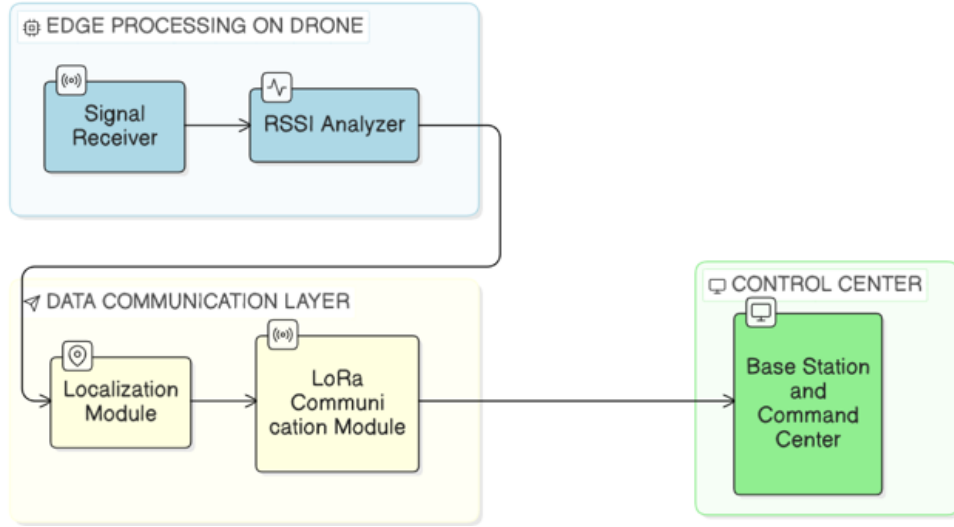


Figure 3.5: Passive Mobile Detection Module

The passive mobile signal detection module detects wireless signals emitted by mobile devices carried by victims. Many smartphones periodically broadcast signals such as Bluetooth or Wi-Fi probe requests that can be detected by nearby receivers.

The system captures these signals and extracts the RSSI values associated with each detected device. RSSI represents the signal strength of the received transmission and can be used to estimate the approximate distance between the drone and the device.

Distance estimation is performed using signal strength analysis, and trilateration techniques are applied to estimate the location of the detected mobile device. This module provides an additional method for detecting victims, particularly in situations where visual detection may not be effective due to obstacles or poor visibility.

3.3.3 LoRa Communication Module

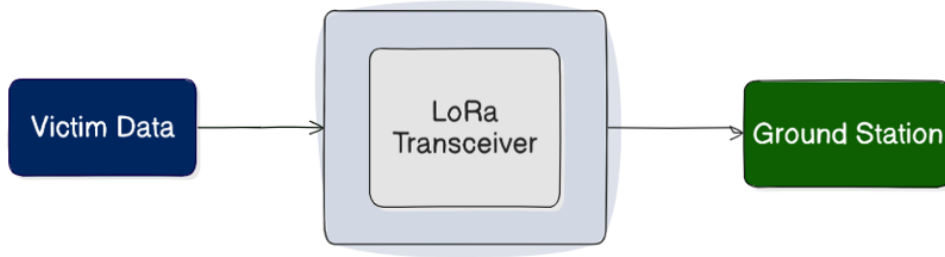


Figure 3.6: LoRa Communication Module

The LoRa communication module enables long-range transmission of victim detection data from the drone to the ground station. LoRa is a low-power wireless communication technology designed for long-distance data transmission with minimal energy consumption.

Once victim detection and localization are completed, the aggregated victim data is transmitted through the LoRa module. The transmitted data packets contain information such as victim coordinates, detection confidence, and system status.

LoRa communication is particularly suitable for disaster environments where conventional communication networks such as cellular or internet connectivity may be unavailable.

3.3.4 Ground Station Module

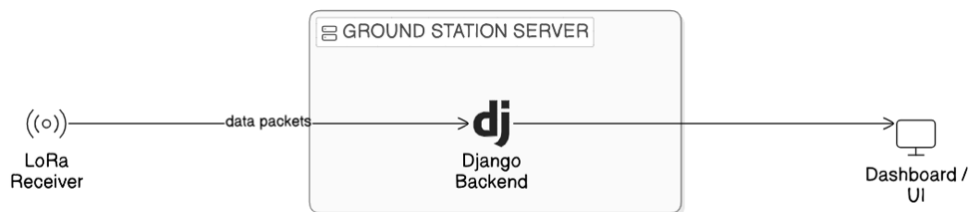


Figure 3.7: Ground Station Module

The ground station module functions as the monitoring and control interface of the disaster response system. It receives victim detection data transmitted from the drone through the LoRa communication link.

A backend server processes the incoming data and stores relevant information. The processed data is then displayed through a dashboard interface that allows operators to visualize victim locations and monitor system activity.

The ground station interface provides situational awareness to rescue teams and supports decision making during rescue operations.

3.4 Tools and Technologies

The proposed system utilizes a combination of hardware components and software tools to implement and evaluate the disaster response framework. The hardware components are used to demonstrate prototype-level functionality, while the software tools support simulation, algorithm development, and system integration.

3.4.1 Hardware Requirements

- Raspberry Pi 5, USB Webcam(Logitech C270), SD Card(64GB), Ublox Neo 6M GPS Module, HC-05 BLE Module, Wi-Fi Adapter, Laptop Intel i5 or equivalent with minimum 8GB RAM and 512GB storage.

3.4.2 Software Requirements

- Windows 10, Raspbian OS, Python, Django, SQLite, Firebase, ArduPilot SITL, MAVProxy, YOLOv5n, OpenCV, RealVNC Viewer, HTML, CSS, JavaScript.

3.5 Work breakdown and Responsibilities

- **Niranjana Vijay K** - Handles drone simulation setup and configuration. Responsible for configuring the drone simulation environment and implementing Wi-Fi signal scanning mechanisms used for passive mobile signal detection. Also contributes to system testing and debugging.
- **Sandra Anna Priyan** - Responsible for the passive mobile signal detection module. Works on simulation of mobile device signals and RSSI-based distance estimation. Contributes to the development of algorithms for signal strength analysis and trilateration-based victim location estimation.

- **Sreelakshmi K S** - Responsible for the simulation of the Edge AI-based victim detection module. Simulated the LoRa-based communication system for transmitting victim location information to the ground station. Contributed to System integration.
- **Teryn Thomas P** - Responsible for the hardware prototype setup and system monitoring components. Handles Raspberry Pi prototype configuration and contributes to the development of the ground station dashboard for monitoring drone status and detected victim locations.
- **All Members** - Collaboratively contribute to requirement analysis, literature review, feasibility study, system architecture design, documentation, and final project presentation.

3.6 Key Deliverables

The project is expected to produce several important outputs during its development. These deliverables represent the major outcomes of the design and development phases of the proposed disaster response system.

- **System Architecture Design:** A detailed architectural model describing the interaction between drones, detection modules, communication systems, and the ground station.
- **Drone Flight Simulation Environment:** A drone simulation setup implemented using ArduPilot SITL and MAVProxy to simulate drone flight behavior and system operation in a disaster response scenario.
- **Edge AI and Passive Detection Simulation:** Simulation of the Edge AI-based human detection module using a YOLO model, tested with webcam and video inputs. The simulation also includes an autonomous passive mobile signal detection framework integrated with LoRa communication and ground station visualization.
- **Hardware Prototype Implementation:** A Raspberry Pi-based prototype demonstrating the integration of detection and signal scanning components.

- **Edge AI Human Detection on Raspberry Pi:** Implementation of YOLO-based human detection on the Raspberry Pi using a webcam for real-time victim identification.
- **Passive Mobile Signal Detection and Localization:** A module implemented for scanning nearby Bluetooth signals emitted by mobile devices and estimating victim proximity using RSSI-based signal strength analysis and trilateration techniques for approximate location estimation.
- **LoRa Communication Framework:** A communication mechanism for transmitting victim detection information from the drone system to the ground station over long distances.
- **Ground Station Dashboard Visualization:** A monitoring dashboard that displays detected victims, signal detections, and location information on a map interface to assist rescue teams.
- **Project Documentation:** Complete documentation including system design, simulation results, prototype implementation details, and the final project report.

3.7 Project Timeline

The project development was carried out over two semesters. The initial stage focused on requirement analysis, literature review, and system design. In the later stage, the simulation environment, detection modules, and prototype components were implemented and tested. The final stage involved system integration, evaluation, and preparation of project documentation.

Task	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar		
Requirement Analysis & Planning										Sandra	
Literature Survey										Niranjana	
Preliminary Analysis & Feasibility Study										Sreelakshmi	
System Architecture Design										Teryn	
Component Design										All	
Drone Simulation Setup and MAVProxy Configuration											
Passive Mobile Detection Simulation											
Edge AI Module Simulation											
Raspberry Pi Prototype Setup											
YOLO Integration on Raspberry Pi											
Bluetooth Signal Scanning Implementation											
RSSI Distance Estimation and Trilateration											
LoRa Communication Simulation											
Dashboard & Visualization Module											
System Integration											
Testing & Debugging											
Documentation & Report Writing											
Final Demo Preparation											

Figure 3.8: Gantt Chart

3.8 Summary of Chapter

This chapter presented the design of the proposed drone-based disaster response system. The overall architecture, component interactions, and functional modules were explained. The chapter also described the technologies used, the responsibilities of project members, and the development timeline. The design provides a structured framework for implementing victim detection, localization, and communication mechanisms to support disaster response operations.

Chapter 4

Implementation Methods

This chapter presents the implementation of the proposed drone-based disaster response system. The system is developed using a combination of hardware prototyping and software simulation to demonstrate victim detection and localization capabilities. The implementation integrates drone simulation, Edge AI-based human detection, passive mobile signal detection, and localization techniques. A Raspberry Pi-based prototype is used to perform image processing and bluetooth signal scanning, while simulation tools are used to model drone operations and communication workflows.

4.1 System Overview

The proposed system is a drone-assisted disaster response framework designed to detect and locate victims in disaster-affected areas. Drone operations are simulated using the ArduPilot SITL environment. Victim detection is performed using a YOLOv8-based object detection model, while passive scanning of Bluetooth signals is used to detect nearby mobile devices and obtain RSSI values. These values are used with a trilateration algorithm to estimate victim location. A Raspberry Pi-based prototype demonstrates edge-level image processing and signal scanning, with results displayed on a ground station interface.

4.2 Hardware Implementation

A hardware prototype was developed to demonstrate the onboard processing capabilities required for a drone-based disaster response system. The prototype is built using a Raspberry Pi which acts as the edge computing unit responsible for processing visual data and scanning bluetooth signals.

A USB webcam is connected to the Raspberry Pi to capture real-time images. These

images are used for human detection using an Edge AI model. The Raspberry Pi processes the camera feed and detects the presence of humans in the captured frames.

In addition to visual detection, the system also performs passive mobile signal detection. A bluetooth module connected to the Raspberry Pi scans for nearby bluetooth signals emitted by mobile devices. The system records the Received Signal Strength Indicator (RSSI) values of detected devices, which are later used for distance estimation.

The hardware prototype demonstrates how lightweight edge computing hardware can perform real-time victim detection and signal scanning. This prototype represents the type of processing unit that could be mounted on an actual drone in a real disaster response scenario.

4.3 Software Implementation

The software implementation includes drone simulation, victim detection algorithms, signal processing, and victim localization techniques.

4.3.1 Drone Simulation

Drone behavior and mission scenarios are simulated using the ArduPilot Software-In-The-Loop (SITL) simulation environment. SITL allows testing of drone navigation and operational workflows without requiring a physical drone.

MAVProxy is used as the ground control interface to interact with the simulated drone, monitor telemetry data, and execute commands. The simulation environment enables testing of the victim detection workflow in a controlled environment.

4.3.2 Edge AI-based Victim Detection

Edge AI-based Victim Detection

Victim detection is implemented using the YOLOv5 object detection model. In the simulation environment, the detection workflow is integrated with the drone simulation to demonstrate how aerial images captured by the drone can be processed for identifying human presence. The model analyzes the input images and generates bounding boxes and confidence scores for detected individuals.

In addition to the simulation environment, the detection module is also implemented on the Raspberry Pi prototype using a webcam for real-time image capture and processing. This demonstrates the feasibility of running the detection algorithm on edge hardware.

4.3.3 Passive Mobile Signal Detection

The system performs passive scanning of bluetooth signals to detect nearby mobile devices carried by victims. The RSSI values of detected signals are collected and used to estimate the distance between the drone and the victim's device.

Passive signal detection provides an additional method of locating victims even when they are not visually visible to the camera. In the simulation environment, mobile signal detection is simulated to demonstrate how the drone can identify nearby devices during search operations. The same concept is implemented in the hardware prototype using a Bluetooth module connected to the Raspberry Pi to scan and record signal strength values from nearby devices.

The approximate location of victims is determined using a trilateration algorithm. Distance values estimated from RSSI measurements are used as inputs for the localization process. By collecting signal readings from multiple drone positions, the system calculates the intersection points of the estimated distances to determine the approximate coordinates of the victim.

4.3.4 Communication Simulation

Communication between the drone and the ground station is conceptually implemented using LoRa-based communication. In this project, LoRa communication is simulated to demonstrate how victim detection data and location information would be transmitted from the drone to the ground control station in a real deployment scenario.

4.4 Summary

The implementation combines hardware prototyping and software simulation to demonstrate the feasibility of the proposed drone-based disaster response system. The Raspberry Pi prototype validates the edge processing capabilities, while the simulation environment enables testing of drone operations, victim detection, and localization algorithms.

Chapter 5

Results and Discussions

5.1 Overview

This chapter presents the results obtained from the implementation of the proposed edge-enabled drone system for disaster response. The system integrates computer vision-based victim detection, passive mobile signal scanning, and wireless communication to support rapid search and rescue operations in disaster environments.

The proposed system uses the YOLOv5n deep learning model deployed on a Raspberry Pi to detect human presence from aerial camera feeds. In addition, a passive mobile signal detection module scans nearby wireless signals and estimates the possible presence of victims carrying mobile devices. The collected information, including victim detection results, signal strength values, and estimated coordinates, is transmitted from the drone to the ground station using a LoRa communication module.

The results presented in this chapter demonstrate the functioning of the system components, including drone simulation, real-time victim detection, signal scanning, and the visualization of victim locations on a monitoring dashboard. The effectiveness of the system is evaluated in terms of detection capability, real-time communication, and system integration for disaster response applications.

5.2 Results

5.2.1 Simulation Results

```
'build' finished successfully (2.813s)
SIM_VEHICLE: Using defaults from (Tools/autotest/default_params/copter.parm)
SIM_VEHICLE: Run ArduCopter
SIM_VEHICLE: "/home/sreelakshmi/ardupilot/Tools/autotest/run_in_terminal_window.sh" "ArduCopter" "/home/sreelakshmi/ardupilot/build/sitl/bin/arducopter" "--model" "+" "--speedup" "1" "--slave" "0" "--defaults" "Tools/autotest/default_params/copter.parm" "--sim-address-127.0.0.1" "-I0" "--home" "12.971600000000002,77.59460000000001,920.0,0.0"
RITW: Starting ArduCopter : /home/sreelakshmi/ardupilot/build/sitl/bin/arducopter --model + --speedup 1 --slave 0 --defaults Tools/autotest/default_params/copter.parm --sim-address-127.0.0.1 -I0 --home 12.971600000000002,77.59460000000001,920.0,0.0
SIM_VEHICLE: Run MavProxy
SIM_VEHICLE: "mavproxy.py" "--retries" "5" "--out" "172.18.192.1:14550" "--master" "tcp:127.0.0.1:5760" "--sitl" "127.0.0.1:5501" "--map" "--console"
RITW: Window access not found, logging to /tmp/ArduCopter.log
Connect tcp:127.0.0.1:5760 source_system=255
Failed to load module: No module named 'console'. Use 'set moddebug 3' in the MAVProxy console to enable traceback
Failed to load module: No module named 'map'. Use 'set moddebug 3' in the MAVProxy console to enable traceback
Log Directory:
Telemetry log: mav.tlog
Waiting for heartbeat from tcp:127.0.0.1:5760
MAV> AP: RCInput: decoding UDP (Pulses)
AP: Calibrating barometer
Detected vehicle 1:1 on link 0
Online system 1
STABILIZE> Mode STABILIZE
AP: ArduCopter V4.7.0-dev (730c7aa5)
AP: fe8642cfdaf14f5eab28e77cd68d9621
AP: Frame: QUAD/PLUS
Received 1363 parameters (ftp)
Saved 1363 parameters to mav.parm
AP: Barometer 1 calibration complete
AP: Barometer 2 calibration complete
Init Gyro**
AP: ArduPilot Ready
AP: AHRS: DCM active
AP: RC7: SaveWaypoint LOW
AP: EKf3 IMU0 initialised
AP: EKf3 IMU1 initialised
AP: AHRS: EKf3 active
AP: EKf3 IMU1 tilt alignment complete
AP: EKf3 IMU0 tilt alignment complete
AP: EKf3 IMU1 MAG0 initial yaw alignment complete
AP: EKf3 IMU0 MAG0 initial yaw alignment complete
AP: GPS 1: probing for u-blox at 230400 baud
AP: GPS 1: detected u-blox
fence present
```

Figure 5.1: SITL

```
[ALERT] Frame 384: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 386: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 388: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 390: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 392: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
Only C and default locale supported with the posix collation implementation
Only C and default locale supported with the posix collation implementation
Case insensitive sorting unsupported in the posix collation implementation
Numeric mode unsupported in the posix collation implementation
[ALERT] Frame 394: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 396: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 398: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 400: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 402: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 404: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 406: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 408: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 410: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 412: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 418: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 420: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 422: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)

=====
EDGE AI MODULE - SESSION SUMMARY
=====
Model: YOLOv8n
Input source: Webcam
Total frames processed: 422
Detection events: 197
Session duration: 34.8 seconds
Detection log: detections/victim_log_20251229_155118.csv
Images saved: detections/images/
=====
(venv) user@user-VivoBook-ASUSLaptop-M3400QA-M3400QA:~/ardupilot$
```

Figure 5.2: Edge AI Module Simulation on Webcam

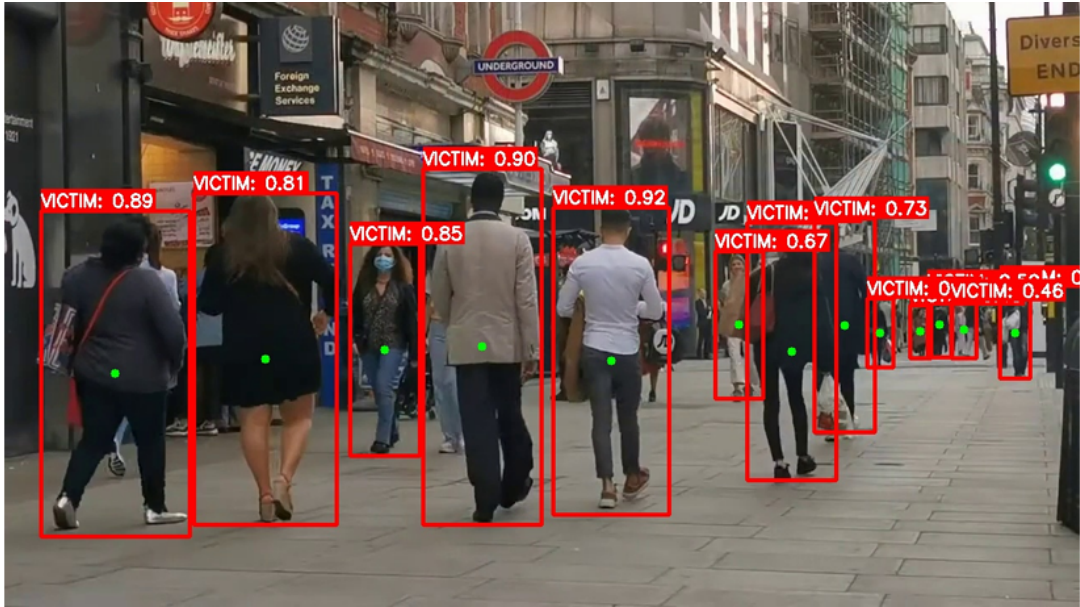


Figure 5.3: Edge AI Simulation on Video

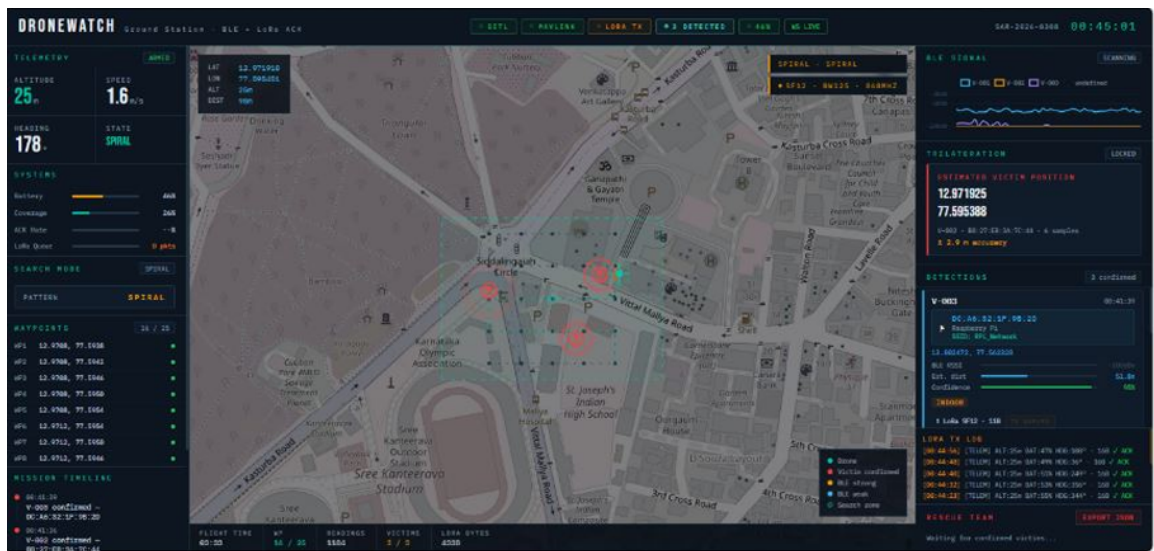


Figure 5.4: Autonomous Passive detection simulation with lora and ground station


```

sreelakshmi@DESKTOP-J35QB71: ~/ardupilot
output add udp:127.0.0.1:14550
STABILIZE> Adding output udp:127.0.0.1:14550
Got COMMAND_ACK: COMPONENT_ARM_DISARM: ACCEPTED
AP: Arming motors
ARMED
Got COMMAND_ACK: DO_SET_MODE: ACCEPTED
GUIDED> Mode GUIDED
Got COMMAND_ACK: NAV_TAKEOFF: ACCEPTED
AP: EKf3 IMU1 MAG0 in-flight yaw alignment complete
AP: EKf3 IMU0 MAG0 in-flight yaw alignment complete
height 15
height 26
Flight battery 100 percent
Flight battery 80 percent
Flight battery 70 percent
Flight battery 50 percent
Flight battery 40 percent
Flight battery 20 percent
Flight battery warning
Got COMMAND_ACK: DO_SET_MODE: ACCEPTED
RTL> Mode RTL
height 15
height 5
AP: SIM Hit ground at 0.499848 m/s
Flight battery 10 percent
Flight battery warning
AP: Disarming motors
DISARMED
Flight battery 10 percent
Flight battery warning

sreelakshmi@DESKTOP-J35QB71: ~/drone_project
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -79dBm true:73.1m est:89.1m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -75dBm true:73.1m est:56.2m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -78dBm true:73.1m est:79.4m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -97dBm true:64.4m est:18.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -76dBm true:73.1m est:63.1m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -80dBm true:73.1m est:100.0m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -78dBm true:73.1m est:79.4m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -100dBm true:87.4m est:51.8m Drone(12.97160,77.59460)
[LORA] TX #47 (16B) TELEM: 02040B000C5ee50049ffff800030521
=====
Tick:880 | Victims:3/3 | Readings:2061 | LoRa:47TX/737B
=====
[RSSI] V-001 MAC:A4:C3:F0:85:2E:11 -81dBm true:73.1m est:112.2m Drone(12.97160,77.59460)
[RSSI] V-002 MAC:B8:27:EB:3A:7C:44 -100dBm true:64.4m est:21.5m Drone(12.97160,77.59460)
[RSSI] V-003 MAC:DC:A6:32:1F:9B:2D -89dBm true:87.4m est:25.1m Drone(12.97160,77.59460)
(yenv) sreelakshmi@DESKTOP-J35QB71: ~/drone_project$

```

Figure 5.5: Autonomous Passive detection simulation with lora and ground station Terminal

5.2.2 Hardware Results

```

C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

D:\final_project\Full code\drone_swarm>activate firstenv

(firstenv) D:\final_project\Full code\drone_swarm>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
March 07, 2026 - 19:16:26
Django version 4.2, using settings 'drone_swarm.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

|

```

Figure 5.6: Terminal for Ground Station

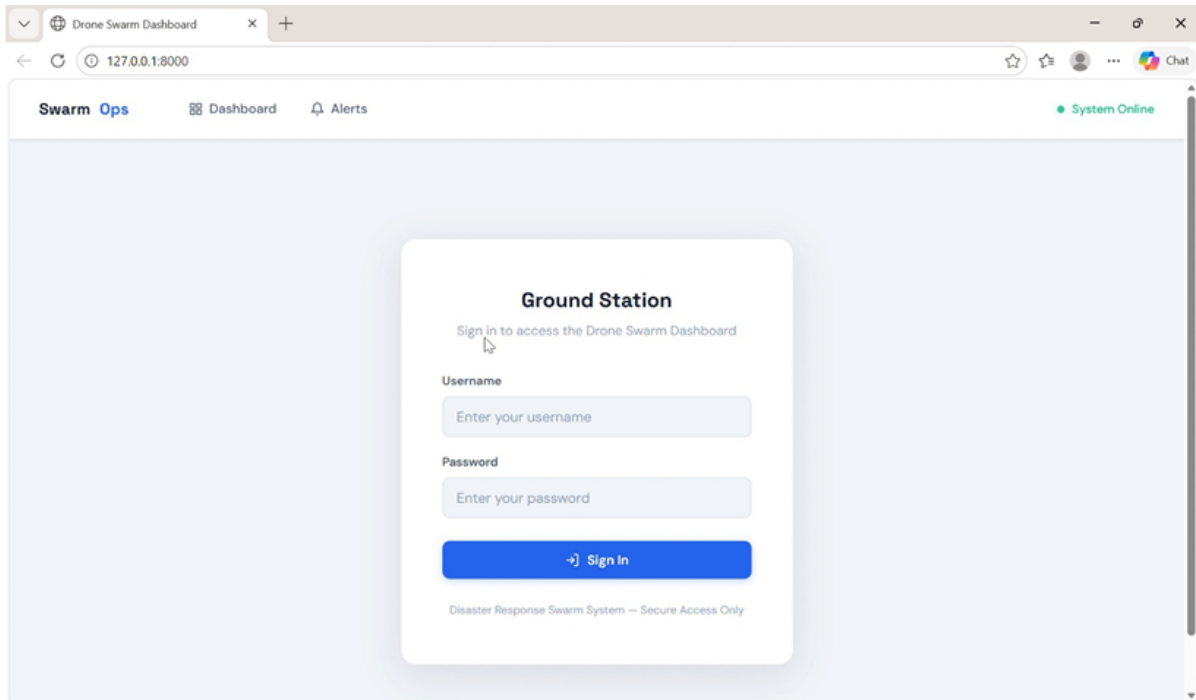


Figure 5.7: Ground Station login page

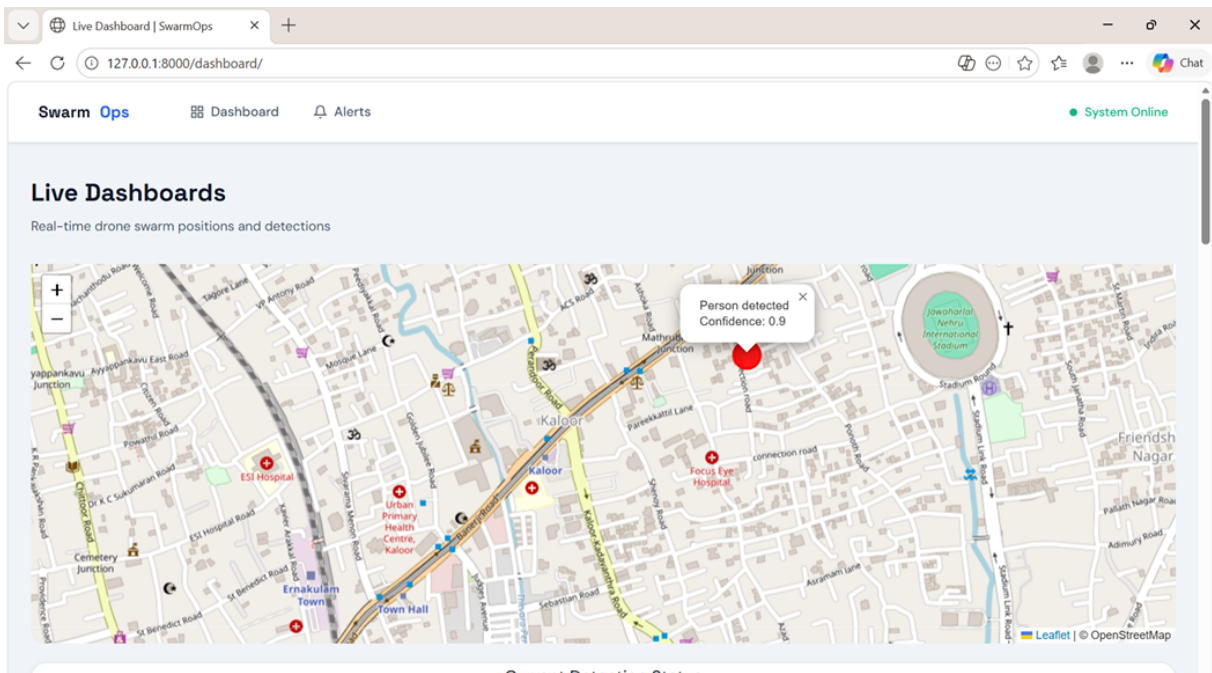


Figure 5.8: Person detected on map with confidence score

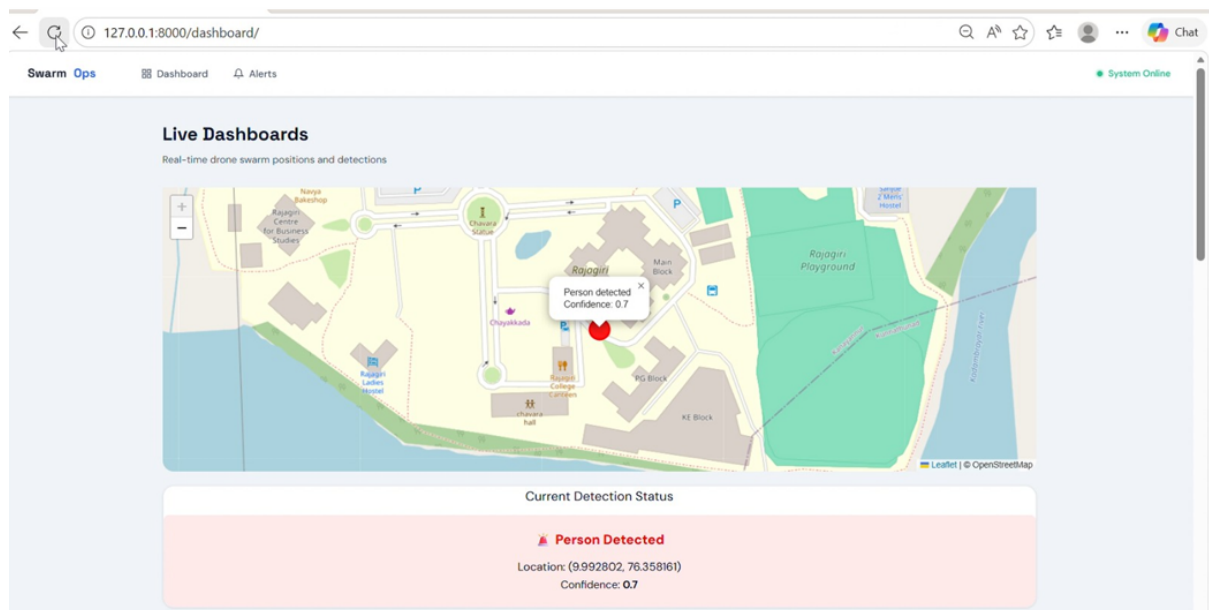


Figure 5.9: Alert with Victim Location and Confidence Score

Live Dashboard | SwarmOps

127.0.0.1:8000/dashboard/

Swarm Ops Dashboard Alerts System Online

Passive Detections

DRONE	RSSI	STATUS	TIME
Drone-1	-64	Signal Found	04 Mar 2026 20:12:20
Drone-1	-64	Signal Found	04 Mar 2026 20:12:18
Drone-1	-64	Signal Found	04 Mar 2026 20:03:19
Drone-1	-64	Signal Found	04 Mar 2026 20:03:15
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-59	Signal Found	04 Mar 2026 19:59:11

Figure 5.10: Passive Detection List

DRONE	PERSON FOUND	COUNT	DISTANCE (M)	TIME
Drone-1	True	1	4.34	09 Mar 2026 16:59:45
Drone-1	True	1	4.34	09 Mar 2026 16:59:45
Drone-1	False	0	None	09 Mar 2026 16:59:44
Drone-1	False	0	None	09 Mar 2026 16:59:44
Drone-1	False	0	None	09 Mar 2026 16:59:43
Drone-1	False	0	None	09 Mar 2026 16:59:43
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42

Figure 5.11: Active Detection(Camera Detection) List

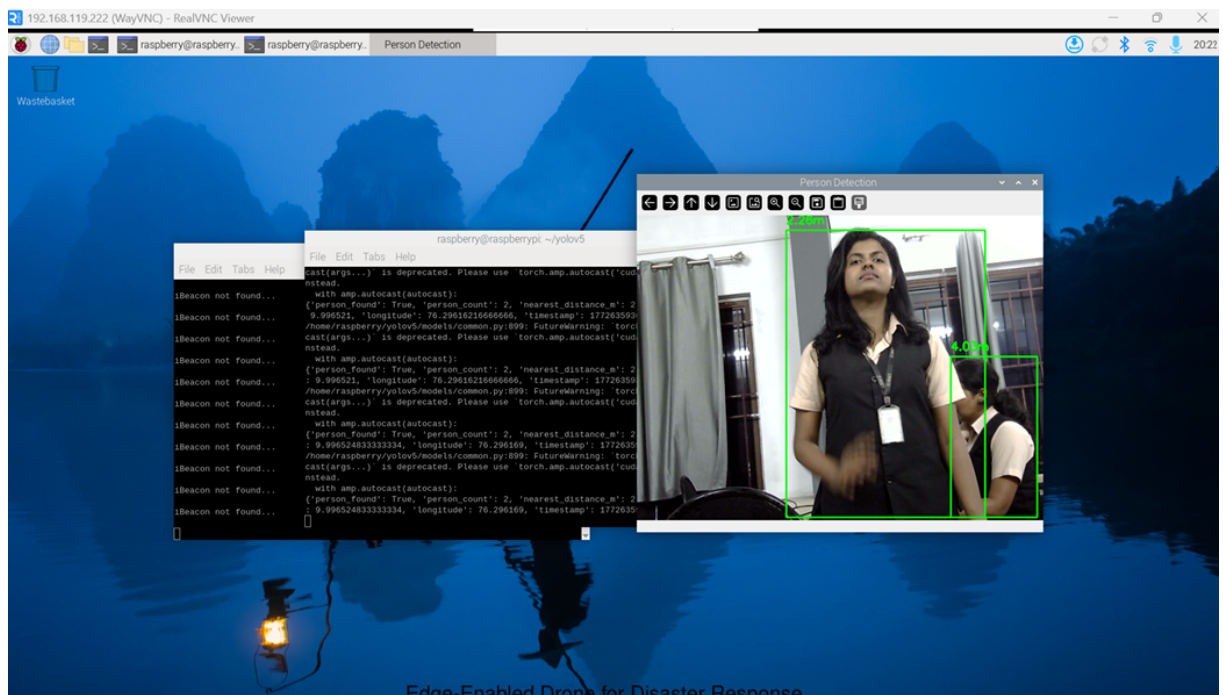


Figure 5.12: YOLO Detection on Raspberry Pi with bounding boxes and approximate distance to victim

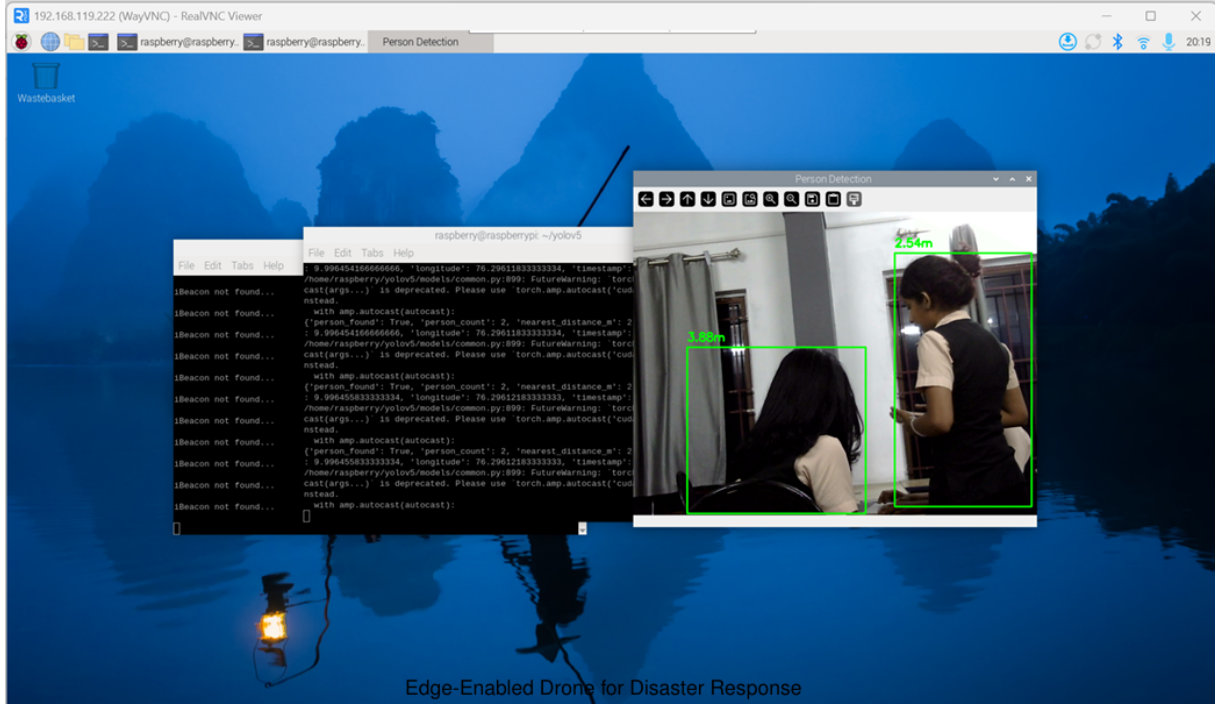


Figure 5.13: YOLO Detection

5.3 Discussion

The results demonstrate that the proposed edge-enabled drone system can effectively support disaster response operations through automated victim detection and communication.

The YOLOv5n model running on the Raspberry Pi enables real-time human detection from aerial camera feeds. The lightweight architecture of the model allows efficient execution on edge devices with limited computational resources, making it suitable for drone-based applications.

The passive mobile signal detection module complements visual detection by identifying nearby mobile devices. This approach increases the chances of detecting victims even when they are partially hidden or not clearly visible in the camera feed.

The use of LoRa communication allows reliable transmission of victim information from the drone to the ground station over long distances with minimal power consumption. This is particularly useful in disaster environments where traditional communication infrastructure may be unavailable.

The real-time dashboard further enhances situational awareness by displaying victim coordinates on a map and maintaining a record of detected victims. This information can

assist rescue teams in planning efficient search and rescue operations.

Although the current implementation demonstrates promising results, future improvements may include the integration of multiple drones, more advanced localization techniques, and deployment in real-world disaster scenarios.

5.4 Summary of Chapter

This chapter presented the results obtained from the implementation of the proposed edge-enabled drone system for disaster response.

The results demonstrate the successful integration of drone simulation, human detection using the YOLOv5n model, passive mobile signal detection, and LoRa-based communication. The system is capable of detecting potential victims, transmitting relevant information to the ground station, and displaying victim locations on a monitoring dashboard.

Overall, the developed system shows strong potential for assisting disaster management teams by enabling faster victim detection and improved situational awareness. Further development and real-world testing can enhance the reliability and practical applicability of the system in large-scale disaster response operations.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

This project presented the design and implementation of an Edge-enabled drone-based disaster response system to assist rescue teams in locating victims in disaster-affected areas. The system integrates drone simulation, Edge AI-based human detection, passive mobile signal detection, RSSI-based localization, and LoRa communication for transmitting victim information to the ground station.

Drone flight behavior was simulated using ArduPilot SITL and MAVProxy, enabling system testing in a controlled environment. The Edge AI module implemented using a YOLO-based model successfully demonstrated real-time human detection using webcam and video inputs. In addition, passive mobile signal detection was implemented by scanning Bluetooth signals from nearby devices. RSSI-based distance estimation and trilateration techniques were used to approximate victim locations.

A Raspberry Pi-based prototype was developed to demonstrate hardware-level implementation of the detection modules. The detected victim information was transmitted through and visualized through a ground station dashboard, where victim locations were displayed on a map interface.

The results demonstrate that combining Edge AI-based visual detection with passive mobile signal detection and long-range communication can improve the reliability of victim identification during disaster response operations.

6.2 Future Scope

Although the proposed system demonstrates the feasibility of integrating multiple detection and communication technologies, several improvements can be considered in future work.

- **Real Drone Deployment:** Implementing the system on an actual drone platform for real-world testing.
- **Improved Localization Accuracy:** Using advanced localization algorithms and sensor fusion techniques to improve victim location estimation.
- **Additional Sensors:** Integrating thermal or infrared sensors to enhance victim detection in low-visibility conditions.
- **Multi-Drone Coordination:** Extending the system to support coordinated operations of multiple drones.
- **Enhanced Communication Network:** Developing a LoRa mesh network for reliable long-range communication.
- **AI Model Optimization:** Optimizing deep learning models for improved performance on edge devices.

These improvements could further enhance the effectiveness of UAV-assisted disaster response systems.

6.3 Summary of Chapter

This chapter summarized the conclusions derived from the implementation and evaluation of the proposed disaster response system. The results demonstrated the feasibility of integrating drone simulation, Edge AI detection, passive mobile signal detection, RSSI-based localization, and LoRa communication within a unified framework. Future enhancements such as real drone deployment, improved localization methods, and multi-drone coordination could further improve the system's capabilities for disaster response operations.

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Appendix A: Presentation

Edge-Enabled Drone for Disaster Response

FINAL PRESENTATION

10/03/2026

Ms. Ann Grace Attokaren

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Contents

- ▶ Problem Definition
- ▶ Problem Objective
- ▶ Purpose and Need
- ▶ Literature Survey
- ▶ Proposed Method
- ▶ Architecture Diagram
- ▶ Module description
- ▶ Results and Outputs
- ▶ Assumptions
- ▶ Work breakdown and Responsibilities
- ▶ Hardware and Software Requirements
- ▶ Gantt Chart
- ▶ Risks and Challenges
- ▶ Conclusion

Problem Definition

- ▶ During disasters such as earthquakes, floods, or landslides, locating victims quickly is extremely difficult due to debris, inaccessible terrain, and damaged infrastructure.
- ▶ Conventional rescue operations rely heavily on manual search methods, which are time-consuming and may delay assistance to victims.
- ▶ Communication networks are often disrupted in disaster zones, making it harder to detect victims or transmit information to rescue teams.

Project Objective

- ▶ To design a drone-based disaster response system capable of detecting and locating victims in disaster-affected areas.
- ▶ To integrate Edge AI-based visual detection and passive mobile signal detection for improved victim identification.
- ▶ To estimate victim location using RSSI-based distance estimation and trilateration techniques.
- ▶ To enable transmission of victim information to the ground station using LoRa-based communication.
- ▶ To develop and demonstrate the system using both drone simulation and a Raspberry Pi-based prototype.

Purpose and Need

- ▶ Rescue operations in disaster areas are often slow due to difficulty in locating victims quickly.
- ▶ There is a need for a system that can assist in identifying and locating victims efficiently.
- ▶ This project aims to improve rescue effectiveness and reduce response time.

Literature Survey

Supporting Search and Rescue Operations with UAVs [1]

Waharte, S., & Trigoni, N., Proc. Int. Conf. on Emerging Security Technologies, IEEE, 2010.

Method

- ▶ Uses UAVs for aerial monitoring of disaster areas.
- ▶ Applies intelligent search strategies and probabilistic models to locate victims.
- ▶ Enables autonomous navigation and real-time data transmission to ground teams.

Advantages

- ▶ Rapid coverage of large and inaccessible areas.
- ▶ Reduces rescue response time.

Disadvantages

- ▶ Limited UAV battery life.
- ▶ Performance affected by environmental conditions.

Literature Survey

Unsupervised Human Detection with Embedded Vision on UAVs for Search and Rescue [2]

Lygouras et al., Sensors, 2019.

Method

- ▶ Uses embedded vision systems on UAVs for human detection.
- ▶ Applies computer vision techniques to analyze aerial images.
- ▶ Performs onboard processing for real-time victim detection.

Advantages

- ▶ Enables automatic victim detection.
- ▶ Reduces dependence on human operators.

Disadvantages

- ▶ Detection accuracy affected by lighting and obstacles.
- ▶ Limited processing power on UAV hardware.

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Literature Survey

Localization of WiFi Devices Using Probe Requests Captured at UAVs [3]

Acuna et al., IEEE WCNC, 2017.

Method

- ▶ UAV captures WiFi probe requests from nearby mobile devices.
- ▶ Uses RSSI measurements to estimate device distance.
- ▶ Applies localization algorithms to estimate device position.

Advantages

- ▶ Detects victims through their mobile devices.
- ▶ Covers large areas quickly using UAVs.

Disadvantages

- ▶ Accuracy affected by signal interference.
- ▶ Requires WiFi-enabled devices.

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Literature Survey

UAV-based Real-Time Survivor Detection System in Post-Disaster Search and Rescue Operations [4]

Dong et al., IEEE Journal on Miniaturization for Air and Space Systems, 2021.

Method

- ▶ Uses UAVs with deep learning models for survivor detection.
- ▶ Applies YOLO-based object detection on aerial images.
- ▶ Provides real-time detection and location information.

Advantages

- ▶ Enables real-time survivor identification.
- ▶ Improves efficiency of search operations.

Disadvantages

- ▶ High computational requirements.
- ▶ Detection affected by occlusion or debris.

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Literature Survey

Edge AI Based Autonomous UAV for Emergency Network Deployment [5]

Biswas et al., WiSPNET, 2022.

Method

- ▶ Deploys Edge AI models on UAVs for survivor detection.
- ▶ Processes image data locally on edge devices.
- ▶ Establishes temporary communication networks in disaster areas.

Advantages

- ▶ Enables low-latency real-time detection.
- ▶ Reduces dependency on cloud connectivity.

Disadvantages

- ▶ Limited computational resources on edge devices.
- ▶ UAV flight duration constrained by battery capacity.

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Literature Survey Comparison

Paper	Technique Used	Key Contribution	Limitations
Supporting Search and Rescue Operations with UAVs (2010)	UAV based aerial monitoring	Uses UAVs to perform aerial surveillance and assist rescue teams in locating victims across large disaster areas	Limited automation and absence of AI based detection
Unsupervised Human Detection with Embedded Vision on UAVs (2019)	Embedded computer vision system	Detects human presence using onboard vision processing directly on the UAV	Detection accuracy affected by lighting conditions and limited onboard processing power
Localization of WiFi Devices Using Probe Requests Captured at UAVs (2017)	RSSI based device localization	Locates victims by capturing WiFi probe requests from mobile devices and estimating their position	Signal interference and environmental factors affect localization accuracy
UAV based Real Time Survivor Detection System (2021)	Deep learning based detection	Uses deep learning models such as YOLO to detect survivors from aerial images in real time	High computational requirements and limited UAV battery life
Edge AI Based Autonomous UAV for Emergency Network Deployment (2022)	Edge AI based detection and communication	Integrates AI based detection, edge processing, and emergency communication networks for disaster response	Limited by edge device processing capability and UAV flight duration

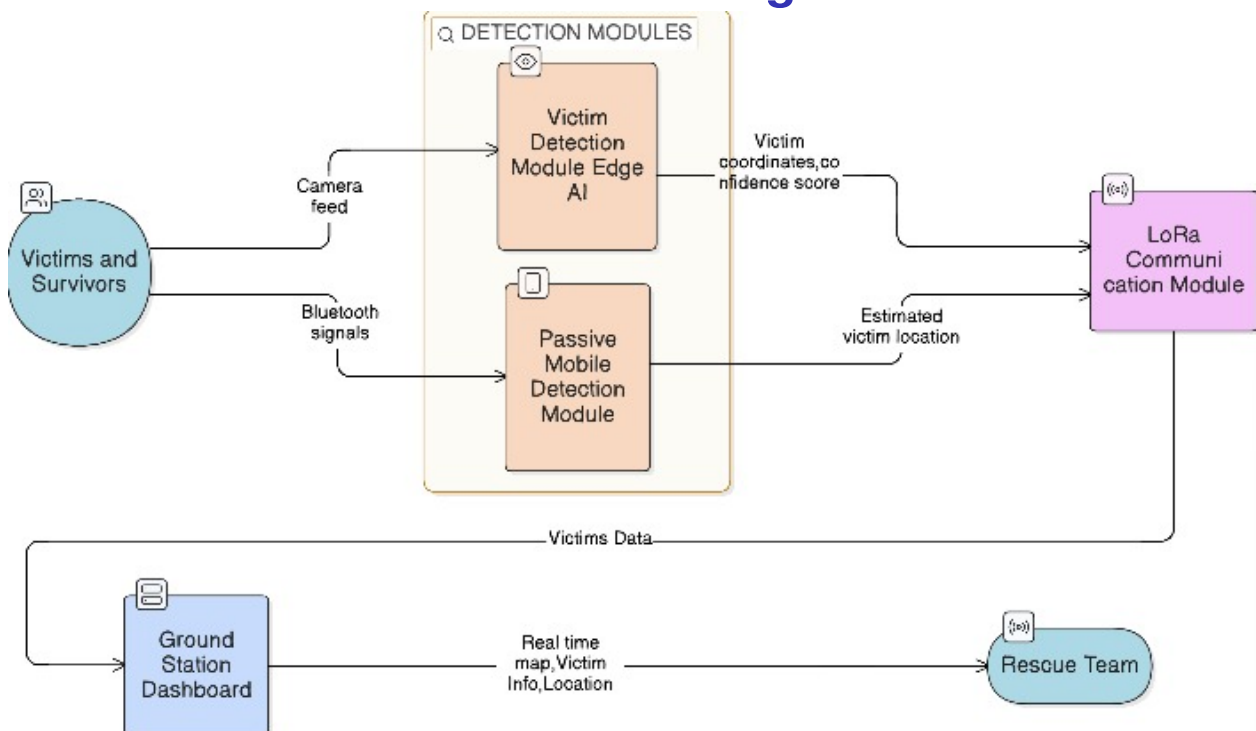
Proposed Method

- ▶ The proposed system is a drone-based disaster response framework designed to detect and locate victims in disaster-affected areas.
- ▶ The system integrates Edge AI-based visual detection and passive mobile signal detection to identify potential victims.
- ▶ Edge AI analyzes aerial images captured by the drone camera to detect human presence.
- ▶ Passive mobile detection scans BLE signals emitted by nearby mobile devices and uses RSSI measurements to estimate the distance to victims.
- ▶ Victim location is estimated using a trilateration algorithm, and detection information is transmitted to the ground station for monitoring and decision making.

Proposed Method

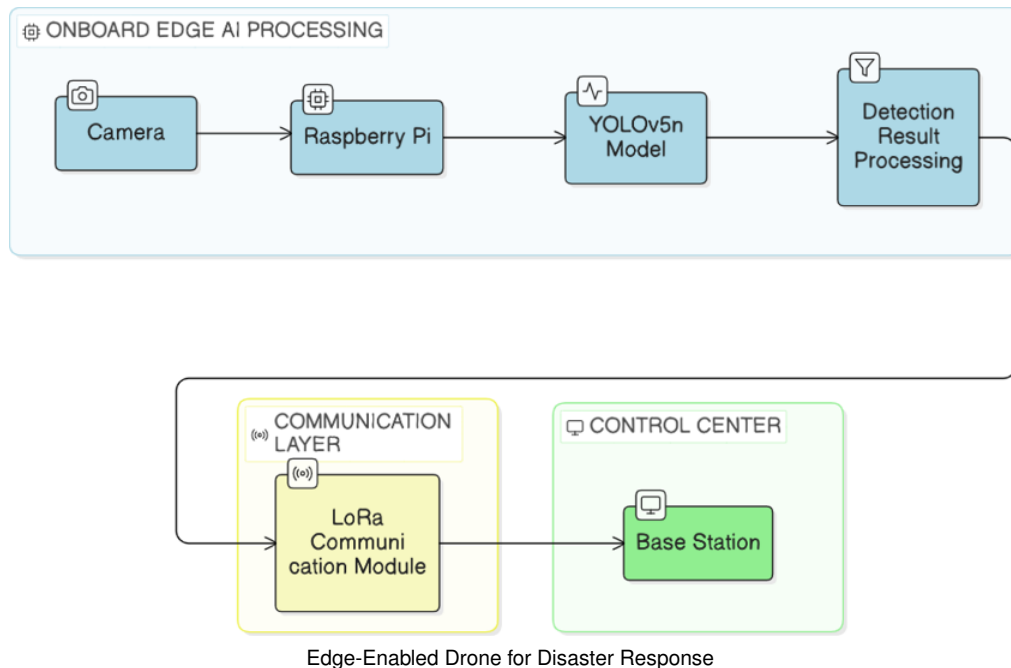
- ▶ The system is implemented using a combination of drone simulation and a Raspberry Pi based hardware prototype.
- ▶ Drone flight and navigation are simulated using ArduPilot SITL and controlled through MAVProxy.
- ▶ The victim detection modules were also simulated to determine feasibility.
- ▶ Edge AI detection and passive signal scanning are demonstrated using a Raspberry Pi with webcam.
- ▶ A ground station interface visualizes victim detection results and location information.

Architecture Diagram



Modules

Edge AI Detection



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Module: Edge AI-Based Victim Detection

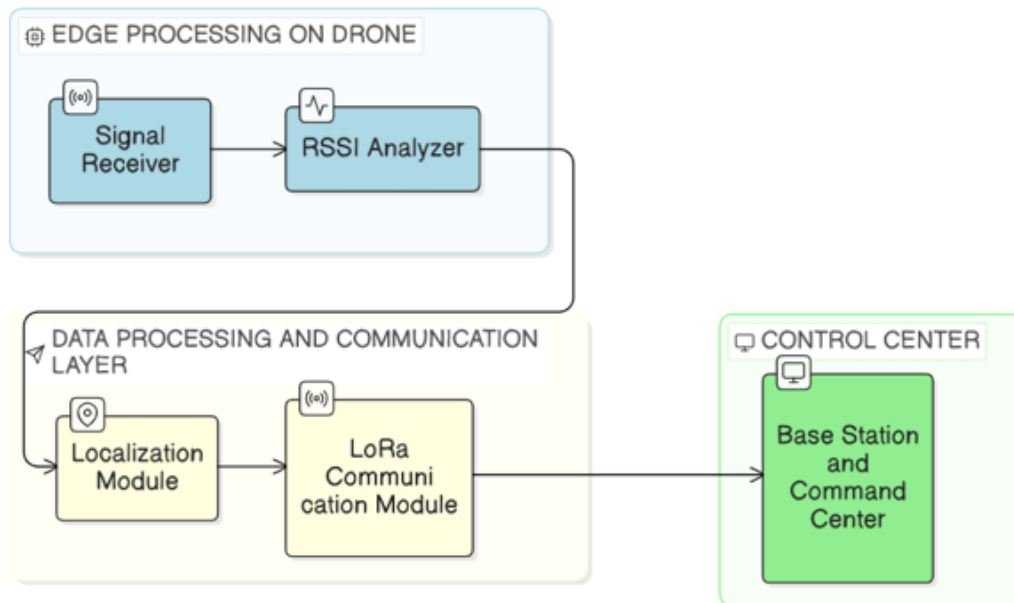
- ▶ In the drone-based system, an onboard camera captures aerial images which are processed using Edge AI to detect victims in real time.
- ▶ A YOLOv5n deep learning model is used to identify human presence and generate detection confidence.
- ▶ Edge processing enables real-time detection without dependence on internet connectivity.
- ▶ **Implementation:** Victim detection is integrated in the ArduPilot SITL drone simulation and implemented physically using a Raspberry Pi and webcam prototype.

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Modules

Passive Mobile Detection



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Module: Passive Mobile Signal Detection

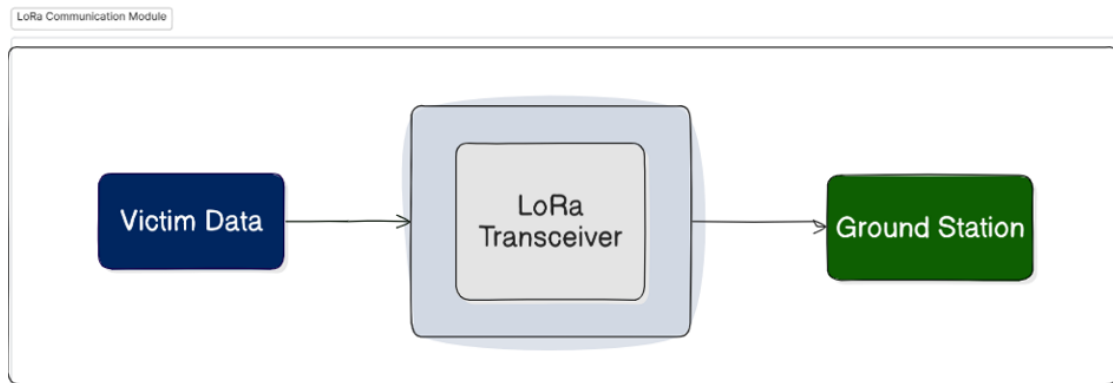
- ▶ The drone detects victims by passively scanning Bluetooth signals emitted from nearby mobile devices.
- ▶ RSSI values are used to estimate the distance between the drone and detected devices.
- ▶ Trilateration algorithm is applied to estimate the victim's location.
- ▶ **Implementation:** Signal detection and localization are simulated in SITL, and demonstrated using a Raspberry Pi for real signal scanning.

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Modules

LoRa Mesh Communication Module

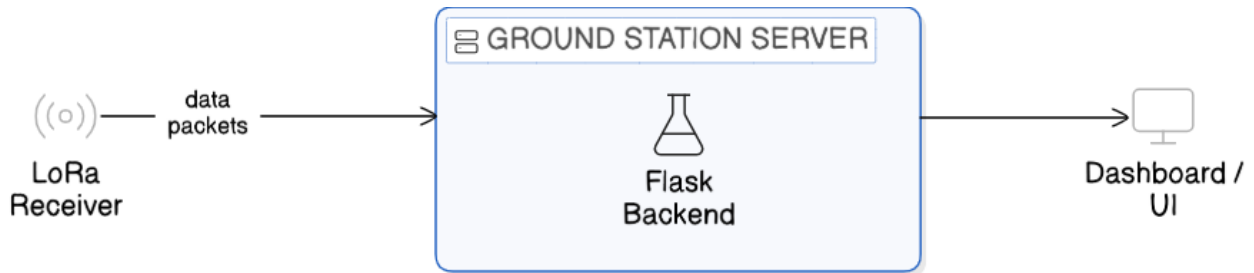


Module: LoRa Communication

- ▶ LoRa enables long-range, low-power communication between the drone and ground station in disaster environments.
- ▶ Detection results and victim coordinates are transmitted to the rescue team.
- ▶ Provides communication capability without relying on cellular or internet infrastructure.
- ▶ **Implementation:** LoRa communication is simulated to demonstrate data transmission between the drone system and ground station.

Modules

Ground Station Module



Module: Ground Station

- ▶ The ground station receives, processes, and displays victim detection information.
- ▶ Provides visualization of victim location and detection status.
- ▶ Enables monitoring and decision-making for rescue operations.
- ▶ **Implementation:** A ground station interface is developed to display data received from Edge AI and Passive mobile detection modules.

Results and Outputs

- ▶ Drone flight simulation using ArduPilot SITL.
- ▶ Simulation of Edge AI-based human detection and Passive mobile signal detection.
- ▶ LoRa communication module is simulated to transmit victim data from drone to ground station.
- ▶ YOLOv5n model successfully detects human presence from webcam feed on Raspberry Pi.
- ▶ Bounding boxes and confidence scores are displayed in real time.

Results

- ▶ Passive scanning captures nearby device signals without user interaction.
- ▶ Victim coordinates, detection confidence scores, and timestamps are successfully transmitted as structured data packets.
- ▶ Real-time dashboard displays victim locations on the map and a list of recent detections.
- ▶ Location coordinates are transmitted to the ground station for real-time victim mapping.

Results

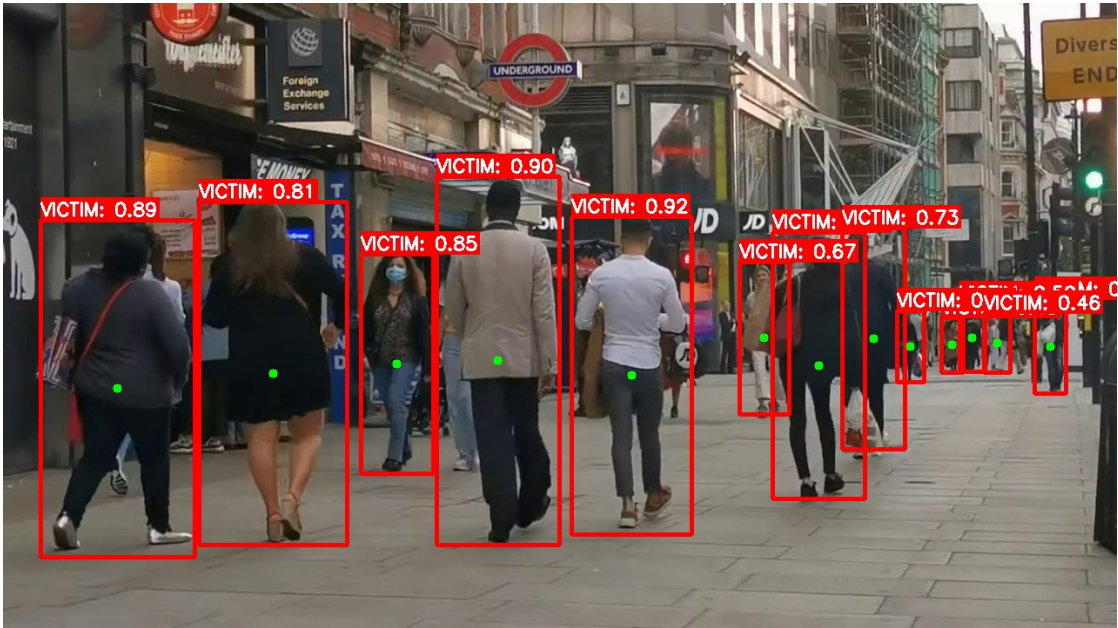
Figure: Edge AI Module Simulation on Webcam

```
[ALERT] Frame 384: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 386: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 388: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 390: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 392: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
Only C and default locale supported with the posix collation implementation
Only C and default locale supported with the posix collation implementation
Case insensitive sorting unsupported in the posix collation implementation
Numeric mode unsupported in the posix collation implementation
[ALERT] Frame 394: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 396: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 398: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 400: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 402: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 404: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 406: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 408: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 410: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 412: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 418: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 420: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)
[ALERT] Frame 422: 1 victim(s)! GPS(-35.363300, 149.164300, 50.0m)

=====
EDGE AI MODULE - SESSION SUMMARY
=====
Model: YOLOv8n
Input source: Webcam
Total frames processed: 422
Detection events: 197
Session duration: 34.8 seconds
Detection log: detections/victim_log_20251229_155118.csv
Images saved: detections/images/
=====
(venv) user@user-VivoBook-ASUSLaptop-M3400QA-M3400QA:~/ardupilot$
```

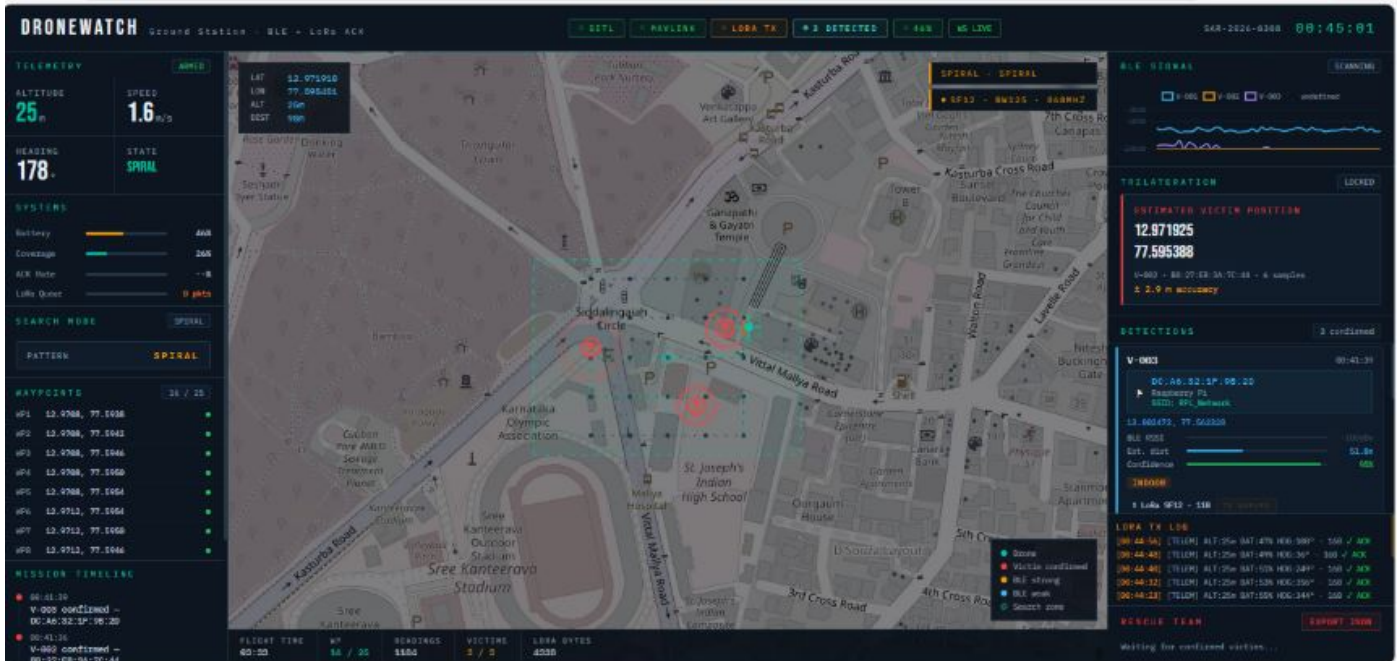
Results

Figure: Edge AI Simulation on Video



Results

Figure:Autonomous Passive detection simulation with lora and ground station

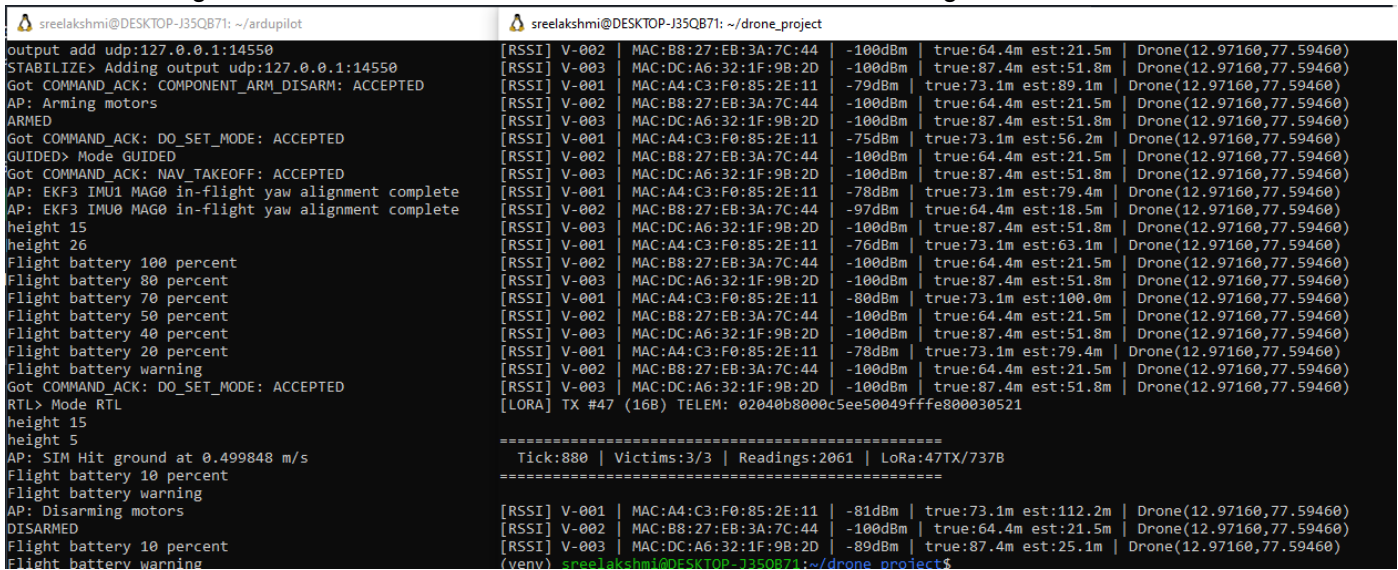


Edge-Enabled Drone for Disaster Response

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Results

Figure:Autonomous Passive detection simulation with lora and ground station Terminal

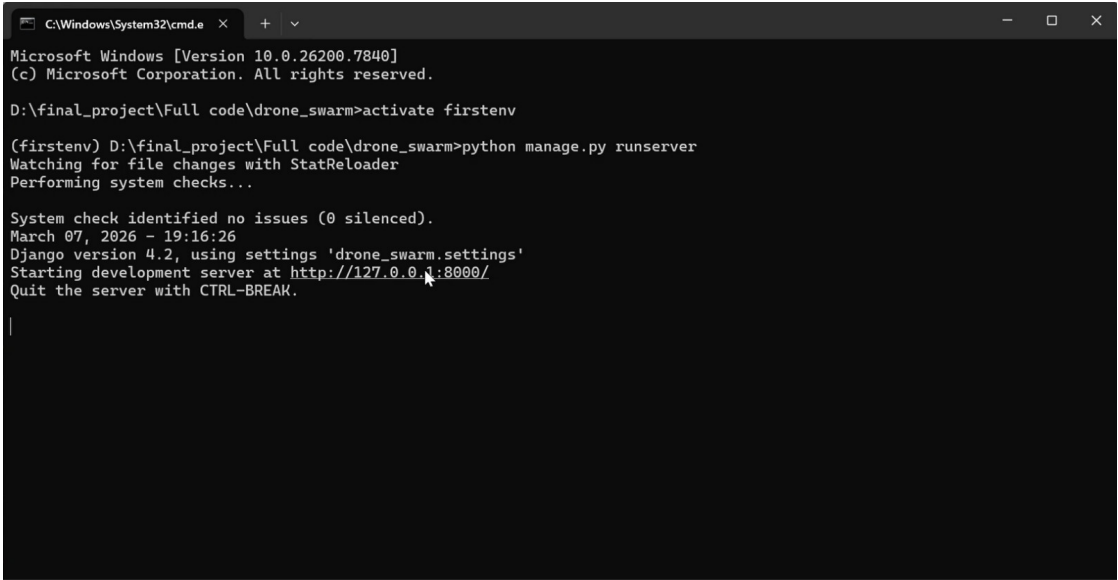


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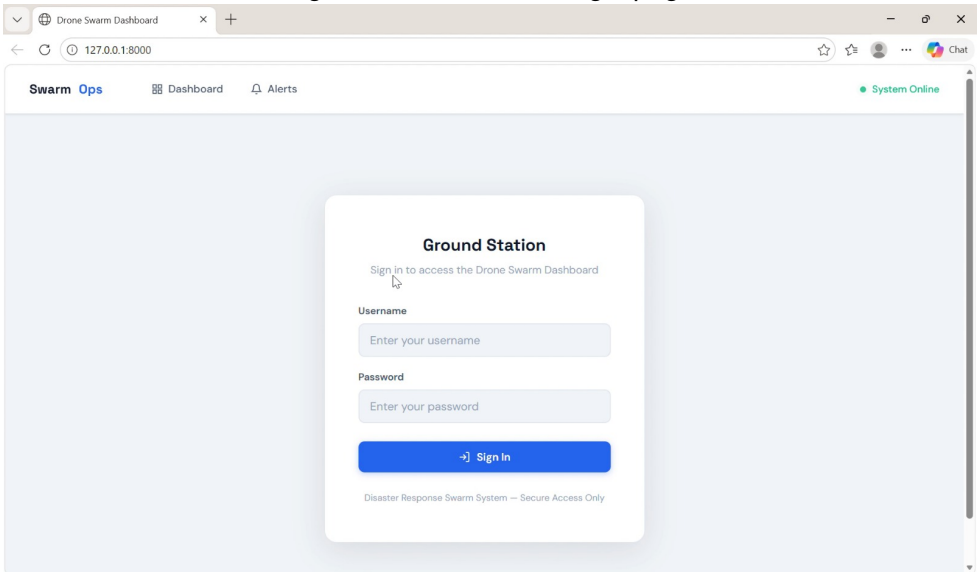
Results

Figure:Terminal for Ground Station



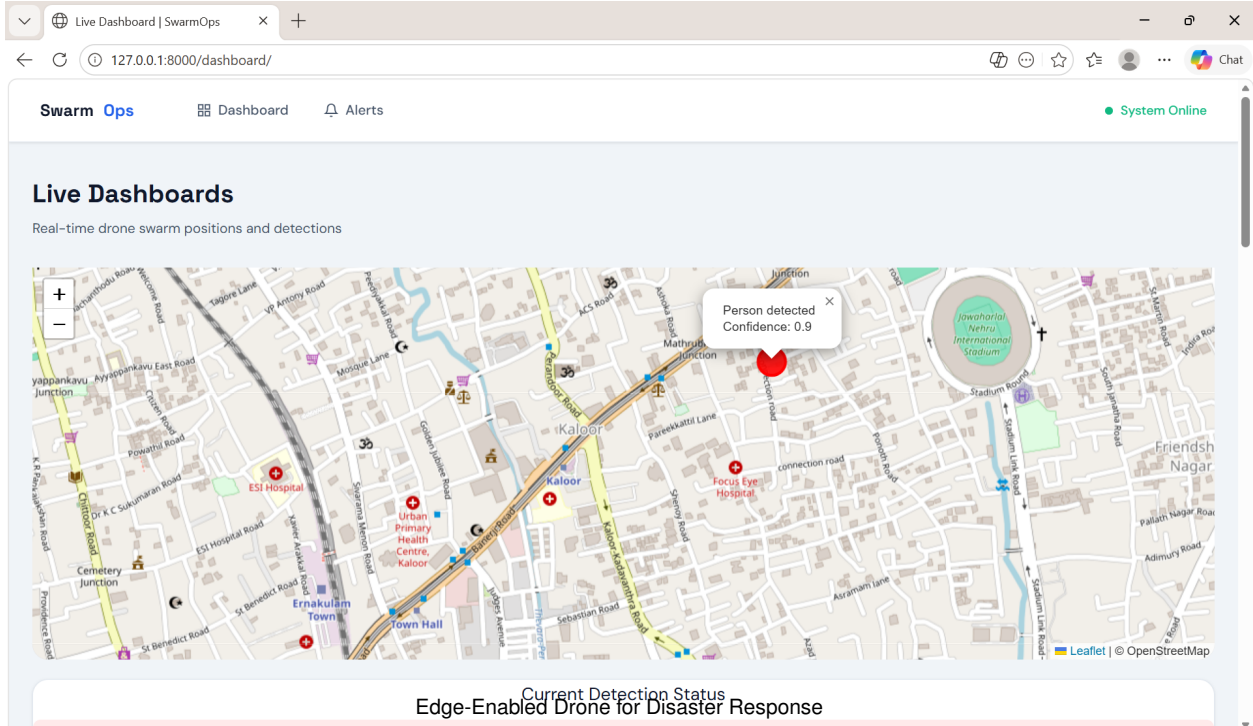
Results

Figure:Ground Station login page



Results

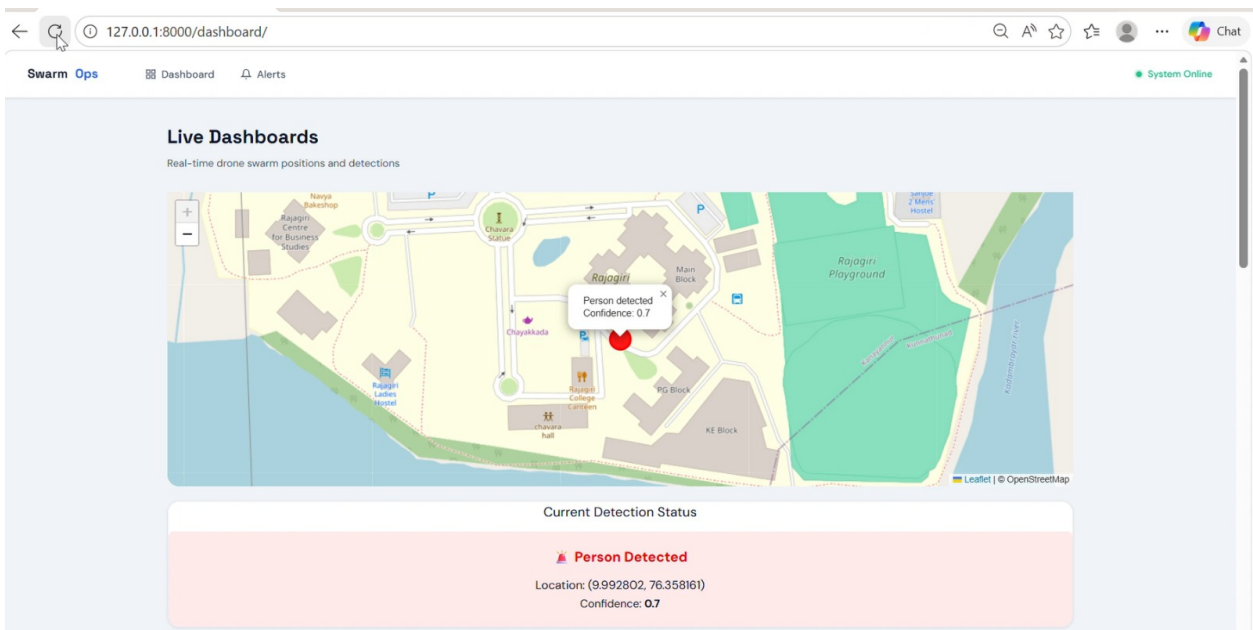
Figure: Person detected on map with confidence score



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Results

Figure: Alert with Victim Location and Confidence Score



Edge-Enabled Drone for Disaster Response

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Results

Figure: Passive Detection List

Live Dashboard | SwarmOps

127.0.0.1:8000/dashboard/

Chat

Swarm Ops

Dashboard Alerts

System Online

Passive Detections			
DRONE	RSSI	STATUS	TIME
Drone-1	-64	Signal Found	04 Mar 2026 20:12:20
Drone-1	-64	Signal Found	04 Mar 2026 20:12:18
Drone-1	-64	Signal Found	04 Mar 2026 20:03:19
Drone-1	-64	Signal Found	04 Mar 2026 20:03:15
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-64	Signal Found	04 Mar 2026 19:59:18
Drone-1	-59	Signal Found	04 Mar 2026 19:59:11
Drone-1	-59	Signal Found	04 Mar 2026 19:59:11

Edge-Enabled Drone for Disaster Response

Results

Figure: Active Detection(Camera Detection) List

127.0.0.1:8000/dashboard/

Summarize Chat

Swarm Ops

Dashboard Alerts

System Online

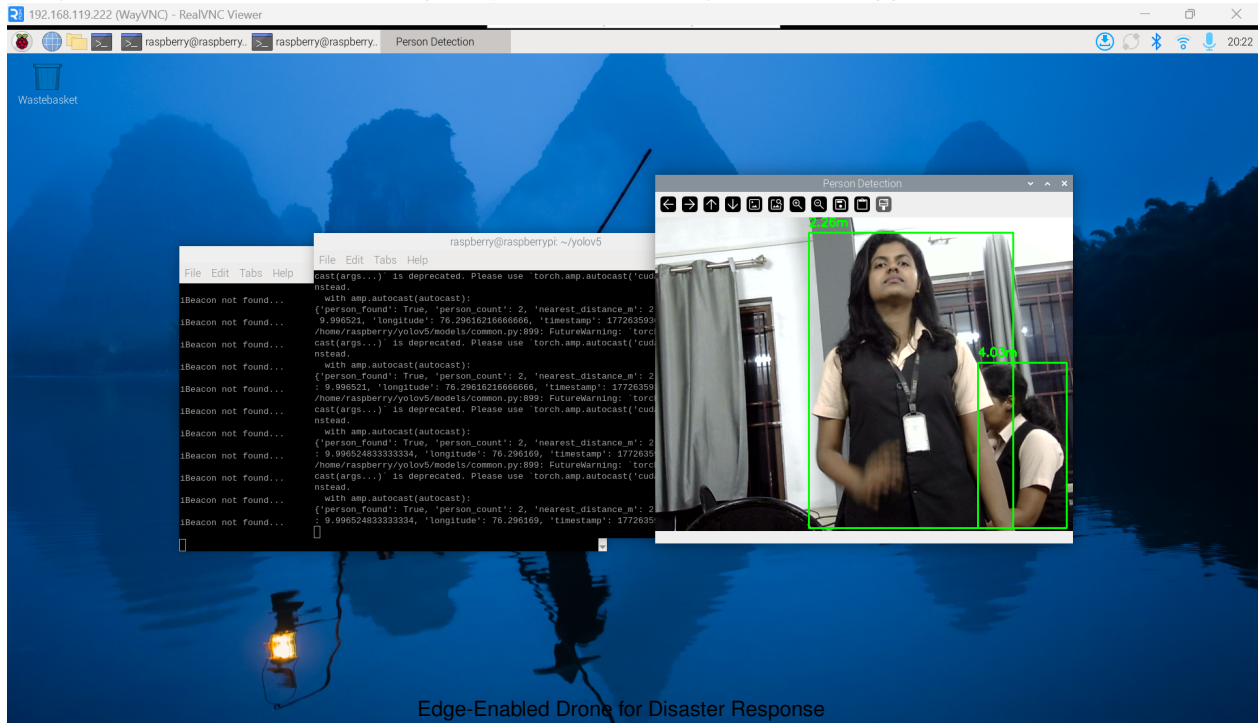
Active Detections				
DRONE	PERSON FOUND	COUNT	DISTANCE (M)	TIME
Drone-1	True	1	4.34	09 Mar 2026 16:59:45
Drone-1	True	1	4.34	09 Mar 2026 16:59:45
Drone-1	False	0	None	09 Mar 2026 16:59:44
Drone-1	False	0	None	09 Mar 2026 16:59:44
Drone-1	False	0	None	09 Mar 2026 16:59:43
Drone-1	False	0	None	09 Mar 2026 16:59:43
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42
Drone-1	False	0	None	09 Mar 2026 16:59:42

Fused Detections

Edge-Enabled Drone for Disaster Response

Results

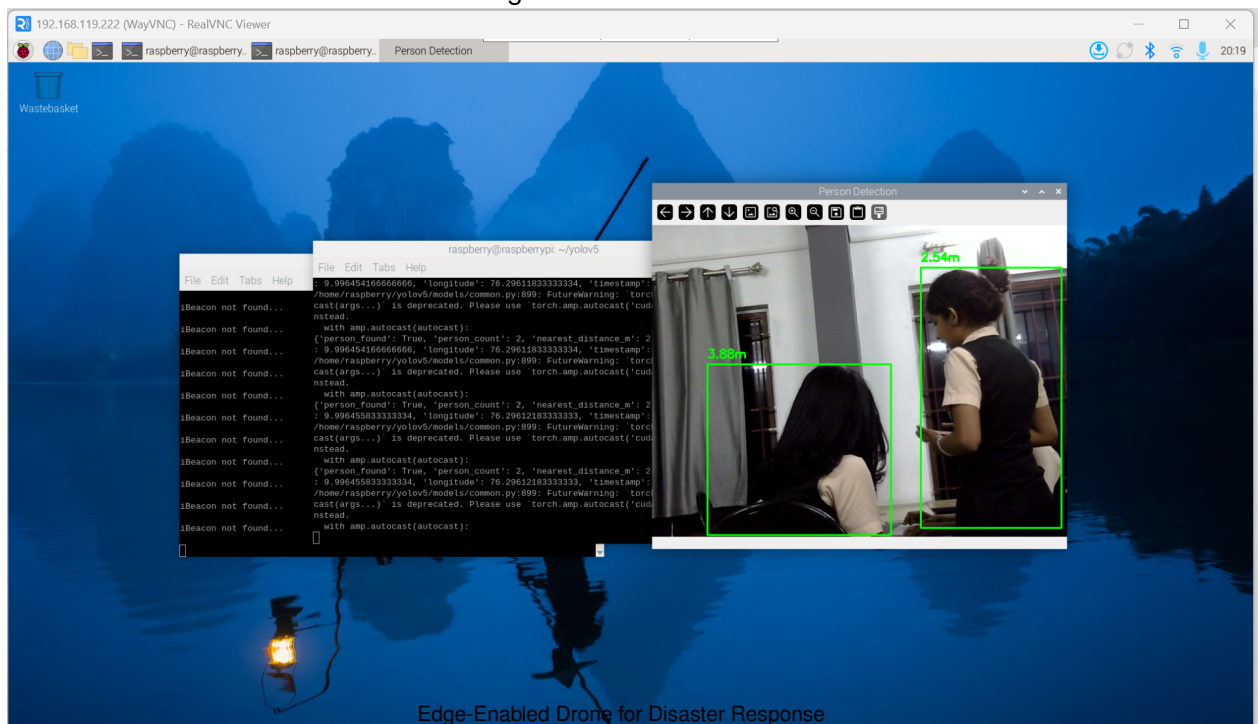
Figure: YOLO Detection on Raspberry Pi with bounding boxes and approximate distance to victim



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Results

Figure: YOLO Detection



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Results

Figure: SITL

```
'build' finished successfully (2.813s)
SIM_VEHICLE: Using defaults from (Tools/autotest/default_params/copter.parm)
SIM_VEHICLE: Run ArduCopter
SIM_VEHICLE: "/home/sreelakshmi/ardupilot/Tools/autotest/run_in_terminal_window.sh" "ArduCopter" "/home/sreelakshmi/ardupilot/build/sitl/bin/arducopter" "--model" "+" "--speedup" "1" "--slave" "0" "--defaults" "Tools/autotest/default_params/copter.parm" "--sim-address-127.0.0.1" "-I0" "--home" "12.971600000000002,77.594600000000001,920.0,0"
RITM: Starting ArduCopter : /home/sreelakshmi/ardupilot/build/sitl/bin/arducopter --model + --speedup 1 --slave 0 --defaults Tools/autotest/default_params/copter.parm --sim-address-127.0.0.1 -I0 --home 12.971600000000002,77.594600000000001,920.0,0.0
SIM_VEHICLE: Run MAVProxy
SIM_VEHICLE: "mavproxy.py" "--retries" "5" "--out" "172.18.192.1:14550" "--master" "tcp:127.0.0.1:5760" "--sitl" "127.0.0.1:5501" "--map" "--console"
RITM: Window access not found, logging to /tmp/ArduCopter.log
Connect tcp:127.0.0.1:5760 source_system=255
Failed to load module: No module named 'console'. Use 'set moddebug 3' in the MAVProxy console to enable traceback
Failed to load module: No module named 'map'. Use 'set moddebug 3' in the MAVProxy console to enable traceback
Log Directory:
Telemetry log: mav.tlog
Waiting for heartbeat from tcp:127.0.0.1:5760
MAV> AP: RCInput: decoding UDP (Pulses)
AP: Calibrating barometer
Detected vehicle 1:1 on link 0
Online system 1
STABILIZE> Mode STABILIZE
AP: ArduCopter V4.7.0-dev (730c7aa5)
AP: fe8642cfdaf14f5eab28e77cd68d9621
AP: Frame: QUAD/PLUS
Received 1363 parameters (ftp)
Saved 1363 parameters to mav.parm
AP: Barometer 1 calibration complete
AP: Barometer 2 calibration complete
Init Gyro++
AP: ArduPilot Ready
AP: AHRS: DCM active
AP: RC7: SaveWaypoint LOW
AP: EKF3 IMU0 initialised
AP: EKF3 IMU1 initialised
AP: AHRS: EKF3 active
AP: EKF3 IMU1 tilt alignment complete
AP: EKF3 IMU0 tilt alignment complete
AP: EKF3 IMU1 MAG0 initial yaw alignment complete
AP: EKF3 IMU0 MAG0 initial yaw alignment complete
AP: GPS 1: probing for u-blox at 230400 baud
AP: GPS 1: detected u-blox
Fence present
```

Assumptions

- ▶ Victims will have detectable signals for identification.
- ▶ RSSI-based distance estimation may be affected by environmental noise and signal multipath effects.
- ▶ Visual detection performance depends on lighting conditions and camera quality.

Work breakdown and responsibilities

Sandra Anna Priyan

- ▶ Passive Mobile Detection Simulation
- ▶ Passive Mobile Signal Scanning on Raspberry Pi
- ▶ RSSI Distance Estimation and Trilateration

Sreelakshmi K S

- ▶ Implemented Edge AI Module Simulation
- ▶ Implemented LoRa communication simulation for victim data transmission
- ▶ System Integration

Work breakdown and responsibilities

Niranjana Vijay K

- ▶ Drone Simulation Setup
- ▶ YOLO Integration on Raspberry Pi
- ▶ Simulation Environment Testing & Debugging

Teryn Thomas P

- ▶ Raspberry Pi Prototype Setup
- ▶ Ground Station Dashboard Development
- ▶ Hardware Prototype Testing and Debugging

Work breakdown and responsibilities

Common Tasks

- ▶ Requirement Analysis and Feasibility Study
- ▶ Literature Survey
- ▶ System Architecture and Module Design
- ▶ Documentation and Demo Preparation

Hardware and Software Requirements

Hardware:

- ▶ Raspberry Pi 5, USB Webcam(Logitech C270), SD Card(64GB), Ublox Neo 6M GPS Module, HC-05 BLE Module, Laptop Intel i5 or equivalent with minimum 8GB RAM and 512GB storage.

Software:

- ▶ Windows 10, Raspbian OS, Python, Django, SQLite, ArduPilot SITL, MAVProxy, YOLOv5n, OpenCV, RealVNC Viewer, HTML, CSS, JavaScript.

Gantt Chart

Task	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar		
Requirement Analysis & Planning										Sandra	
Literature Survey										Niranjana	
Preliminary Analysis & Feasibility Study										Sreelakshmi	
System Architecture Design										Teryn	
Component Design										All	
Drone Simulation Setup and MAVProxy Configuration											
Passive Mobile Detection Simulation											
Edge AI Module Simulation											
Raspberry Pi Prototype Setup											
YOLO Integration on Raspberry Pi											
Bluetooth Signal Scanning Implementation											
RSSI Distance Estimation and Trilateration											
LoRa Communication Simulation											
Dashboard & Visualization Module											
System Integration											
Testing & Debugging											
Documentation & Report Writing											
Final Demo Preparation											

Edge-Enabled Drone for Disaster Response

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Risks and Challenges

- ▶ RSSI-based localization accuracy may be affected by signal noise, interference, and multipath propagation.
- ▶ Visual victim detection performance depends on lighting conditions, camera quality, and environmental factors.
- ▶ Simulation may not fully replicate real-world drone dynamics and environmental conditions.
- ▶ Limited processing power of Raspberry Pi may affect real-time performance.
- ▶ GPS signals are typically reliable only in open outdoor areas.

Edge-Enabled Drone for Disaster Response

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Conclusion

- ▶ An edge-enabled drone based framework for disaster response is presented, integrating simulation based validation with prototype level hardware consideration.
- ▶ Edge AI-based visual detection, passive mobile signal detection, and LoRa-based communication enable reliable victim detection and localization in connectivity deprived environments.
- ▶ Simulation results demonstrate the feasibility of the proposed detection and localization approach, providing a foundation for hardware implementation.
- ▶ Prototype implementation validates the feasibility of integrating edge AI, passive signal detection, and real-time ground station visualization for disaster response.
- ▶ The proposed architecture is scalable, cost-effective, and suitable for deployment in both urban and remote disaster scenarios.

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-long Learning: Recognize the need for, and have the preparation and ability for (i) independent and life-long learning (ii) adaptability to new and emerging technologies and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying stan-

dard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.

Mapping	Level	Justification
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.

Mapping	Level	Justification
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.

Mapping	Level	Justification
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.

Mapping	Level	Justification
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.